
Supplementary Material: Sanity Checks and Reproducibility Discipline

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Conn2Conn Preprint Draft

1 Supplementary Material: Sanity Checks and Reproducibility Discipline

This document is a long-form supplement for the Conn2Conn preprint. It combines the compact manuscript supplement with the detailed sanity-check notes from the repository, plus selected machine-readable CSV summaries. The purpose is to make the negative claims auditable: reduction choices, FC measurement noise, nonlinear capacity, richer tractography, family objectives, and operational grid completeness are all checked explicitly.

2 Supplementary Material

This supplement documents the reproducibility and sanity-check discipline behind the main paper. The purpose is not only to provide extra numbers, but to show how the analysis avoids common silent failure modes: population-mean-dominated metrics, split drift, missing grid cells, leakage, dimensionality artifacts, unstable scalar estimators, measurement-noise explanations, and objective mismatch.

2.1 S1. Data, Parcellations, and Frozen Splits

All analyses used HCP Young Adults subjects with both resting-state FC and diffusion-derived SC, plus demographics and anatomy-derived brain-volume features. Connectomes were vectorized as the upper triangle of each parcellation-specific connectivity matrix.

Two parcellations were used:

Parcellation	Regions	Edges	Notes
Glasser	360	64,620	HCP-MMP1.0 multimodal cortical atlas
4S456Parcels	456	103,740	400 cortical Schaefer-style parcels plus 56 subcortical/cerebellar regions

Splits were frozen once and reused everywhere. The split contract is central to the paper: reconstruction, imputation artifact generation, downstream prediction, leak checks, and family-mechanism analyses all load the same `seed{0..9}.json` split files. Splits are family-aware: related subjects are kept on the same side of the train/test boundary. The same subject sets are available across Glasser and 4S456Parcels, so the parcellation comparison is within-cohort rather than confounded by different subjects.

Source artifacts:

- `reproduction/freeze_splits.py`
- `reproduction/splits/seed{0..9}.json`
- `reproduction/_grid_common.py`

Preprint.

2.2 S2. Metrics and Estimators

2.2.1 S2.1 Why demeaned_pearson is primary

Raw connectome correlations are dominated by the group-mean connectome. This makes raw Pearson large even when individual deviations are not recovered. The primary reconstruction metric is therefore `demeaned_pearson`: subtract the training-set mean target connectome, then compare each subject's predicted and observed deviation.

The main text leads with `demeaned_pearson` and uses fingerprinting metrics (`avg_rank`, `top1_acc`) as secondary checks. Conventional MSE, R2, and raw Pearson are retained in the grid for diagnostic purposes but should not drive interpretation.

2.2.2 S2.2 Estimator reporting rule

Every number should be reported with its estimator.

Reconstruction rows include:

- PCA(256)->PLS(64)->inverse-PCA (`pca_pls`)
- BayesianRidge on PCA components (`bayesian_ridge`)
- KernelRidge RBF 3x3 gamma/alpha sweep (`kernel_ridge`)

Downstream scalar rows should be reported with BayesianRidge. The grid includes PCA->PLS and KernelRidge scalar rows for completeness, but these are diagnostic-only. A resolved discrepancy showed why: the older report's cognition baseline and oracle numbers matched BayesianRidge exactly, while an analyst had accidentally read PCA->PLS rows. For scalar targets, PCA->PLS and KernelRidge produced ill-conditioned behavior, including extremely negative R2 and parcellation-inconsistent results for parcellation-independent inputs.

Key resolved examples:

Quantity	PCA->PLS	BayesianRidge
Glasser SC->SC oracle, <code>demeaned_pearson</code>	0.376	0.648
Glasser bv+demo -> CogTotal, Pearson	0.129	0.359

Source artifact:

- `reproduction/exploration/DISCREPANCY_RESOLUTION.md`

2.3 S3. Grid Completeness and Leak Guardrails

The reproduction grid is designed around an explicit completeness contract. Expected cells are generated from the grid configuration before running, and completed outputs are checked against that manifest. Missing cells, unexpected non-finite values, and silently absent metrics are treated as failures rather than tolerated as blank table entries.

The spine grid contains 13,640 cells:

- 2,640 reconstruction cells
- 11,000 downstream cells

The merged output is complete: 13,640/13,640 cells, zero non-finite values in load-bearing metrics, and zero hard leak failures.

Leak checks separate expected biological signal from implementation leakage. Sex and age prediction from connectomes can be real biological signal, while sex/age prediction from incorrectly derived inputs can indicate leakage. The grid records verdict categories:

Verdict	Meaning
ok	No threshold violation
EXPECTED_SIGNAL	Raw connectome predicts sex/age; interpreted as biological signal, not a leak
EXEMPT_FLAGGED	Input contains bv+demo; cognition rows are interpretable but sex/age is expected

Verdict	Meaning
LEAK_FAIL	Non-exempt derived input exceeds leak threshold

Observed counts:

Verdict	Count
ok	3344
EXPECTED_SIGNAL	4
EXEMPT_FLAGGED	1052
LEAK_FAIL	0

Only the combined observed FC+SC input crossed the sex threshold as expected biological signal. No demographic-free derived input triggered LEAK_FAIL.

Source artifacts:

- reproduction/gen_expected_cells.py
- reproduction/configs/expected_cells.csv
- reproduction/verify_completeness.py
- reproduction/outputs/leak_verdict.csv
- reproduction/reports/reproduction_findings.md

2.4 S4. Reduction-Axis Robustness

A natural concern is that the FC->SC > SC->FC asymmetry could be produced by the chosen PCA reduction rather than the data. The reduction-axis check tested five dimensionality strategies over 10 seeds and both directions:

Method	Median FC->SC dp	Median SC->FC dp	Median ratio	p vs 1
PCA->PLS->PCA	0.1355	0.0847	1.621x	0.001
Full PLS on raw 64,620 edges	0.1382	0.0762	1.813x	0.001
JL Gaussian dense	0.0837	0.0610	1.397x	0.001
JL sparse auto	0.0841	0.0592	1.389x	0.001
JL sparse 1/3	0.0846	0.0535	1.554x	0.001

All methods preserve the asymmetry. Full PLS, which removes learned PCA reduction entirely, yields an even stronger ratio. Random Johnson-Lindenstrauss projections lower absolute reconstruction quality, as expected, but retain the direction. The asymmetry is therefore a data property rather than a PCA artifact.

Source artifact:

- notebooks-FC_to_SC-experimental/sanity_checks/preprocessing_check/findings.md

2.5 S5. Reliability Ceilings and FC Measurement Noise

The main question for SC->FC is whether the weak prediction reflects true cross-modal independence or merely noisy FC targets. HCP-YA includes repeated FC measurements, allowing an FC-side reliability analysis.

2.5.1 S5.1 FC is noisy by edge but reliable as a whole connectome

At the single-edge level, much of the between-subject variance is noise. Yet the whole connectome is highly identifiable.

Parcellation	Trait	State/day	Within-session	Noise	G of averaged connectome
Glasser	0.301	0.035	0.021	0.643	0.585
4S456	0.261	0.035	0.021	0.683	0.519

Whole-connectome fingerprint top1 was 0.927 in Glasser and 0.934 in 4S456, with discriminability near 1.0. The reconciliation is that individual signal is distributed: edges are noisy, but the multivariate pattern is stable.

2.5.2 S5.2 FC reproducibility ceiling

Between-session FC reliability is the valid FC reproducibility ceiling. In Glasser, REST1 vs REST2 reached $demeaned_r=0.491$, fingerprint $top1=0.933$, and $avg_rank=0.992$. The within-session LR/RL rung was lower because phase-encode distortion differs across directions, so it is not used as the clean short-interval ceiling.

Parcellation	Comparison	demeaned_r	Pearson	top1	avg_rank
Glasser	within-session LR/RL	0.364	0.710	0.780	0.968
Glasser	between-session REST1/REST2	0.491	0.813	0.933	0.992
4S456	within-session LR/RL	0.337	0.670	0.780	0.969
4S456	between-session REST1/REST2	0.457	0.783	0.936	0.993

2.5.3 S5.3 SC->FC captures a small fraction of reproducible FC

Using the Glasser between-session ceiling of 0.491, SC->FC captures about 17% of reproducible FC in $demeaned_pearson$ and about 5% of the fingerprinting ceiling.

Source -> FC	Metric	Achieved	Ceiling	Fraction
SC->FC	demeaned_r	0.085	0.491	0.17
SC->FC	top1	0.051	0.933	0.05
bv+demo->FC	demeaned_r	0.098	0.491	0.20

2.5.4 S5.4 Per-subject reliability does not explain SC->FC

Per-subject FC reliability is heterogeneous, spanning roughly 0 to 0.78. If SC->FC were limited by FC measurement noise, cleaner subjects should be more predictable from SC. They are not. A subject's SC->FC quality is uncorrelated with that subject's FC reliability (Pearson $r=0.01$, $p=0.72$). Filtering to high-reliability subjects raises the ceiling but leaves SC->FC prediction near 0.08, reducing the fraction captured.

The correct caveat is that this is FC-side evidence. HCP-YA does not provide SC test-retest dMRI, so a uniform SC noise floor cannot be fully ruled out. The supported conclusion is that the SC->FC gap is not explained by FC measurement noise or per-subject FC reliability.

Source artifact:

- notebooks-FC_to_SC-experimental/sanity_checks/noise_sanity_check/findings_noise.md

2.6 S6. Nonlinear, Residual, Sink, and Scaling Nulls

The nonlinear sanity checks ask whether the linear ceiling is a modeling failure. They use several orthogonal probes:

Check	Question	Interpretation
Model-class nonlinearity	Do KernelRidge/HGB recover hidden reconstruction or cognition signal?	No stable gain over linear baselines
Residual boost	If the model is handed the linear solution, can it learn the residual?	No meaningful residual signal
Cross-modal sink	Do FC x SC x r2t interactions unlock cognition?	No useful downstream gain
Scaling curve	Does a nonlinear gap grow from $n=100$ to $n=683$?	Gap is flat rather than expanding

KernelRidge also proved hyperparameter-insensitive in the reproduction grid: the 3x3 gamma/alpha sweep had within-seed max-min spreads around 0.004-0.005 in $demeaned_pearson$. The nine variants therefore behave almost like a single estimator for the present data.

Source artifacts:

- notebooks-FC_to_SC-experimental/non-linear-sanity-check/findings_nonlinear.md
- notebooks-FC_to_SC-experimental/non-linear-sanity-check/findings_residual.md
- notebooks-FC_to_SC-experimental/non-linear-sanity-check/findings_scaling.md
- notebooks-FC_to_SC-experimental/non-linear-sanity-check/nonlinear_synthesis.csv
- notebooks-FC_to_SC-experimental/non-linear-sanity-check/residual_synthesis.csv
- notebooks-FC_to_SC-experimental/non-linear-sanity-check/scaling_synthesis.csv

2.7 S7. Richer Structural Representations

The tractography extension asks whether streamline-count SC is too crude. Named-bundle tractography (r2t) was tested as an alternative structural representation.

The result was negative. r2t features predicted FC worse than count-SC, added no marginal value over SC in marginal tests, and did not lift cognition above the bv+demo floor. This supports the interpretation that the missing cognition signal is not simply hiding in a richer structural feature set, at least in this regime.

Source artifacts:

- notebooks-FC_to_SC-experimental/tractography_predict/e1_source_rep_results.csv
- notebooks-FC_to_SC-experimental/tractography_predict/e2_asymmetry_summary.csv
- notebooks-FC_to_SC-experimental/tractography_predict/e3_marginal_summary.csv
- notebooks-FC_to_SC-experimental/tractography_predict/e5_downstream_summary.csv

2.8 S8. Family Signal and Objective Tradeoff

The family-mechanism grid ports the exploratory notebook analyses into the same frozen-split, both-parcellation framework used by the main grid. The ports are validated against notebook outputs with exact or near-exact regression tests.

Main findings:

Finding	Result	Interpretation
Predicted SC carries family signal	pred_SC_resid_bvdemo sibling AUC 0.810 Glasser / 0.816 4S456	FC-predicted SC contains heritable information
Baseline is weak	bv+demo sibling AUC 0.563 / 0.565	Family signal is not reducible to the baseline
Combined reconstruction objective collapses family ID	combined_pred_SC AUC 0.505 / 0.501	Reconstruction and identification are different
Raw predicted SC separates partly	pred_SC_raw AUC 0.680 / 0.670	Some family signal survives before residualization

This is the constructive counterweight to the cognition result. Predicted connectomes can be useful, but not generically. The task objective determines which signal survives.

Source artifacts:

- reproduction/family_mechanism/outputs/family_auc.csv
- reproduction/family_mechanism/tests/test_aggregation_matches_notebook.py
- reproduction/family_mechanism/tests/test_f8_localization_matches_notebook.py

2.9 S9. Mechanism Mode, Confounds, and Localization

The mechanism analysis should be framed as exploratory. The strongest version is not "PC3 is the mechanism"; it is "a property-selected, low-variance, FC-predictable SC mode carries weak family signal." The component index migrates across parcellations: PC3 in Glasser and PC4 in 4S456.

Important safeguards:

- PC1 is strongly confounded with sex and brain volume (R2 about 0.89-0.93) and should not be interpreted as a family mechanism.
- An earlier PC2 signal failed under cross-seed alignment, illustrating why single-seed mechanisms are fragile.
- The mechanism mode has small effects: FC-predictability around R2=0.22-0.24 and sibling AUC around 0.58.

- The spatial localization concentrates in visual and dorsal-attention edges, but the interpretation is hedged.

The tractography-reliability check tested whether this localization was simply due to high-strength, short-distance, more reliable tractography edges. Strength and distance explained about 41% of |PC3| magnitude, so the confound is real. After partialling both, the visual/DAN enrichment survived:

Network pair	Raw enrichment	Residualized enrichment
visual-visual	11.97x	13.32x
dorsal attention-dorsal attention	7.73x	5.73x
dorsal attention-visual	4.49x	4.99x

Thus reliability-related proxies explain part of the magnitude but do not erase the localization. The rigorous retest-ICC version remains future work.

Source artifacts:

- reproduction/family_mechanism/outputs/f8_per_pc.csv
- reproduction/family_mechanism/outputs/f8_stability.csv
- reproduction/family_mechanism/outputs/f8_pc3_localization.csv
- reproduction/family_mechanism/outputs/f8_pc3_enrichment_agg.csv
- notebooks-FC_to_SC-experimental/sanity_checks/tract_check/findings.md

2.10 S10. Operational Reproducibility

The project uses a two-layer design:

1. Exploratory notebooks discovered the findings and mechanism hypotheses.
2. The reproduction/ suite imports the same shared setup functions, freezes the splits, reruns confirmatory grids on both parcellations and 10 seeds, writes CSV mirrors, and verifies completeness.

CSV outputs are the source of truth. W&B is a replay layer, not a dependency. This matters because offline or failed syncs cannot change the analysis.

The operational discipline includes:

- Git-synced compute state.
- Frozen split manifests.
- CSV append with provenance.
- Expected-cell manifest before running.
- Verification after merging.
- Leak verdicts separated from scientific outcomes.
- Notebook-vs-grid regression checks for family and mechanism analyses.

Source artifacts:

- dev-notes/PROJECT-HANDOFF-2026-06-22.md
- planning/reproducibility_and_grid_plan_theory.md
- planning/reproducibility_and_grid_plan_runlog.md
- reproduction/upload_to_wandb.py

3 Machine-Readable Summary Tables

3.0.1 Reduction-axis summary

method	j_l_variant	n_seeds	median_ratio	min_ratio	max_ratio	wilcoxon_p_vs_1	median_FC_to_SC_dp	median_SC_to_FC_dp
FULL_PLS		10	1.813	1.493	2.115	0.0009766	0.1382	0.07616
JL_PLS_PCA	gaussian_dense	10	1.397	1.194	2.25	0.0009766	0.08372	0.06102
JL_PLS_PCA	sparse_auto	10	1.389	1.109	1.903	0.0009766	0.08411	0.05918
JL_PLS_PCA	sparse_third	10	1.554	1.235	1.776	0.0009766	0.08458	0.05348

method	j1_variant	n_seeds	median_ratio	min_ratio	max_ratio	wilcoxon_p_vs_1	median_FC_to_SC_dp	median_SC_to_FC_dp
PCA_PLS_PCA		10	1.621	1.324	1.861	0.0009766	0.1355	0.0847

Full source: notebooks-FC_to_SC-experimental/ sanity_checks/ preprocessing_check/ reduction_axis_summary.csv.

3.0.2 Reduction-axis all seed ratios

method	j1_variant	seed	FC->SC	SC->FC	ratio
FULL_PLS		0	0.1381	0.077	1.793
FULL_PLS		1	0.1381	0.07533	1.833
FULL_PLS		2	0.1323	0.0829	1.596
FULL_PLS		3	0.1383	0.08175	1.692
FULL_PLS		4	0.1371	0.06864	1.997
FULL_PLS		5	0.1476	0.07143	2.066
FULL_PLS		6	0.1446	0.0684	2.115
FULL_PLS		7	0.1394	0.08403	1.659
FULL_PLS		8	0.1403	0.07292	1.924
FULL_PLS		9	0.1375	0.09206	1.493
JL_PLS_PCA	gaussian_dense	0	0.07473	0.04884	1.53
JL_PLS_PCA	gaussian_dense	1	0.08485	0.06164	1.377
JL_PLS_PCA	gaussian_dense	2	0.07312	0.06125	1.194
JL_PLS_PCA	gaussian_dense	3	0.08504	0.06212	1.369
JL_PLS_PCA	gaussian_dense	4	0.08259	0.06079	1.359
JL_PLS_PCA	gaussian_dense	5	0.09936	0.05904	1.683
JL_PLS_PCA	gaussian_dense	6	0.0936	0.04161	2.25
JL_PLS_PCA	gaussian_dense	7	0.08232	0.05476	1.503
JL_PLS_PCA	gaussian_dense	8	0.09086	0.06412	1.417
JL_PLS_PCA	gaussian_dense	9	0.08192	0.06181	1.325
JL_PLS_PCA	sparse_auto	0	0.08423	0.05982	1.408
JL_PLS_PCA	sparse_auto	1	0.08235	0.06101	1.35
JL_PLS_PCA	sparse_auto	2	0.06875	0.05659	1.215
JL_PLS_PCA	sparse_auto	3	0.08399	0.06127	1.371

Showing first 24 of 50 rows. Full source: notebooks-FC_to_SC-experimental/ sanity_checks/ preprocessing_check/ reduction_axis_synthesis.csv.

3.0.3 FC reliability ceiling

parc	n	comparison	demeaned_pearson	pearson	top1_acc	avg_rank	mse	r2
Glasser	1018	within_session_run1	0.3645	0.7099	0.775	0.9682	0.03312	-0.38
Glasser	1018	within_session_run2	0.3598	0.7005	0.8291	0.9888	0.03304	-0.3432
Glasser	1018	between_session	0.4905	0.8134	0.9332	0.9916	0.01792	-0.1195
4S456Parcels	1018	within_session_run1	0.3371	0.6704	0.775	0.9694	0.03239	-0.4662
4S456Parcels	1018	within_session_run2	0.331	0.6609	0.8261	0.9896	0.03222	-0.4263
4S456Parcels	1018	between_session	0.4565	0.7832	0.9361	0.9934	0.01748	-0.2337

Full source: notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ a_reliability_ceiling.csv.

3.0.4 FC per-subject reliability summary

parc	n	mean	median	std	min	max	p1	p5	p25	p75	p95	p99	skew	kurtosis	non_gaussian
Glasser	1018	0.4894	0.4943	0.1173	-0.001942	0.7751	0.149	0.2818	0.4204	0.5722	0.6645	0.7138	-0.5807	0.7912	5.2
4S456Parcels	1018	0.4554	0.4624	0.1106	0.05831	0.7451	0.1633	0.2583	0.3904	0.5321	0.6197	0.6727	-0.4183	0.2862	1.8

Full source: notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ g_per_subject_summary.csv.

3.0.5 Reliability filtering summary

source	filter	n_kept	mean_achieved	mean_ceiling	fraction_of_ceiling
SC	all	857	0.08318	0.4914	0.1693
SC	drop_rel_below_0.2	841	0.08309	0.4983	0.1667
SC	drop_bottom_10pct	771	0.08258	0.5171	0.1597
SC	keep_top_50pct	429	0.07953	0.5824	0.1366
bv+demo	all	857	0.102	0.4914	0.2076
bv+demo	drop_rel_below_0.2	841	0.1019	0.4983	0.2045
bv+demo	drop_bottom_10pct	771	0.104	0.5171	0.2012
bv+demo	keep_top_50pct	429	0.1147	0.5824	0.197

Full source: notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ h_reliability_filtered_summary.csv.

3.0.6 Noise synthesis

headline
FC between-session reliability ceiling (Glasser): demeaned_r=0.491, fingerprint top1=0.933, avg_rank=0.992
FC edge variance (Glasser): trait=30%, state=4%, within-session=2%, noise=64%; averaged-connectome reliability G=0.59
SC->FC captures 17% of the reproducible FC signal (demeaned_r 0.085 of ceiling 0.491)

Full source: notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ noise_synthesis.csv.

3.0.7 Tractography source representation

rep	seed	target	method	mse	r2	pearson	avg_rank	top1_acc	demeaned_pearson
SC	0	FC	shared_pca	0.01398	-0.06216	0.8307	0.7125	0.03077	0.0849
r2t	0	FC	shared_pca	0.01506	-0.1403	0.8176	0.6082	0.01538	0.03969
r2t_corr	0	FC	shared_pca	0.0145	-0.09805	0.8242	0.5935	0.02051	0.03549
SC_r2t	0	FC	block_pca	0.01416	-0.07531	0.8286	0.6948	0.01538	0.07302
kitchen_sink	0	FC	block_pca	0.01406	-0.06776	0.8295	0.7082	0.03077	0.0853
SC	1	FC	shared_pca	0.01445	-0.05236	0.8224	0.7213	0.05128	0.08621
r2t	1	FC	shared_pca	0.01534	-0.114	0.8115	0.6277	0.01538	0.05158
r2t_corr	1	FC	shared_pca	0.01516	-0.09946	0.8135	0.5857	0.005128	0.02439
SC_r2t	1	FC	block_pca	0.01449	-0.05483	0.8219	0.7238	0.02564	0.08606
kitchen_sink	1	FC	block_pca	0.01433	-0.04436	0.8238	0.7554	0.06154	0.1059
SC	2	FC	shared_pca	0.01432	-0.05061	0.8277	0.7082	0.05128	0.08343
r2t	2	FC	shared_pca	0.01545	-0.1304	0.8146	0.602	0.01538	0.04751
r2t_corr	2	FC	shared_pca	0.01479	-0.08391	0.8223	0.6145	0.01026	0.04052
SC_r2t	2	FC	block_pca	0.01447	-0.06154	0.8263	0.7045	0.02051	0.07893

Showing first 14 of 50 rows. Full source: notebooks-FC_to_SC-experimental/ tractography_predict/ e1_source_rep_results.csv.

3.0.8 Tractography asymmetry summary

rep	metric	comparison_kind	n_seeds	median_FCwins	min_FCwins	max_FCwins	wilcoxon_p_FCwins	median_FC_to_X	median_Y
SC	demeaned_pearson	ratio_FCwins	10	1.621	1.324	1.861	0.0009766	0.1355	0.0847
SC	pearson	ratio_FCwins	10	1.101	1.096	1.111	0.0009766	0.9125	0.8292
SC	top1_acc	ratio_FCwins	10	2.562	1.5	5	0.0009766	0.1179	0.05128
SC	avg_rank	ratio_FCwins	10	1.225	1.174	1.265	0.0009766	0.8787	0.7128
SC	mse	ratio_FCwins	10	2.594	2.518	2.639	0.0009766	0.005533	0.01429
SC	r2	diff_FCwins	10	0.03595	0.01707	0.05097	0.0009766	-0.02205	-0.05716
r2t	demeaned_pearson	ratio_FCwins	10	2.264	1.871	2.83	0.0009766	0.1113	0.04925
r2t	pearson	ratio_FCwins	10	1.037	1.029	1.04	0.0009766	0.8453	0.8153
r2t	top1_acc	ratio_FCwins	9	1.5	0.6667	3	0.03906	0.02821	0.02051
r2t	avg_rank	ratio_FCwins	10	1.185	1.121	1.232	0.0009766	0.7212	0.609
r2t	mse	ratio_FCwins	10	2.835e-08	2.819e-08	2.909e-08	1	5.365e+05	0.01537
r2t	r2	diff_FCwins	10	0.4459	-0.3524	0.7425	0.001953	0.3234	-0.127
r2t_corr	demeaned_pearson	ratio_FCwins	10	1.634	1.178	2.39	0.0009766	0.05816	0.03625
r2t_corr	pearson	ratio_FCwins	10	0.9813	0.9751	0.9898	1	0.8071	0.8231

rep	metric	comparison_kind	n_seeds	median_FCwins	min_FCwins	max_FCwins	wilcoxon_p_FCwins	median_FC_to_X	median
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Showing first 14 of 18 rows. Full source: notebooks-FC_to_SC-experimental/ tractography_predict/ e2_asymmetry_summary.csv.

3.0.9 Tractography marginal summary

n_seeds	median_dp_SC	median_dp_SC_r2t	median_delta	min_delta	max_delta	wilcoxon_p_two_sided	wilcoxon_p_one_sided_greater
10	0.0847	0.07908	-0.001289	-0.01188	0.0061	0.04883	0.9814

Full source: notebooks-FC_to_SC-experimental/ tractography_predict/ e3_marginal_summary.csv.

3.0.10 Tractography downstream summary

rep	target	pearson_raw	pearson_resid	lift_over_bvdemo_raw
FC	CogCrystalComp_Unadj	0.4508	0.3737	0.1046
FC	CogFluidComp_Unadj	0.3423	0.2179	0.03286
FC	CogTotalComp_Unadj	0.4508	0.2636	0.07845
SC	CogCrystalComp_Unadj	0.2674	0.08626	-0.07878
SC	CogFluidComp_Unadj	0.1798	0.006016	-0.1297
SC	CogTotalComp_Unadj	0.2614	0.04037	-0.1109
SC_r2t	CogCrystalComp_Unadj	0.1597	0.02804	-0.1865
SC_r2t	CogFluidComp_Unadj	0.1352	0.07167	-0.1743
SC_r2t	CogTotalComp_Unadj	0.1958	0.04758	-0.1765
bv+demo	CogCrystalComp_Unadj	0.3462		0
bv+demo	CogFluidComp_Unadj	0.3095		0
bv+demo	CogTotalComp_Unadj	0.3723		0
r2t	CogCrystalComp_Unadj	0.1636	0.02728	-0.1826
r2t	CogFluidComp_Unadj	0.1303	0.06082	-0.1792

Showing first 14 of 21 rows. Full source: notebooks-FC_to_SC-experimental/ tractography_predict/ e5_downstream_summary.csv.

3.0.11 Nonlinear cognition summary

rep	estimator	target	pearson	spearman	r2	lift_over_bvdemo
FC	HGB	CogCrystalComp_Unadj	0.2493	0.2184	0.0541	-0.117
FC	HGB	CogFluidComp_Unadj	0.118	0.1419	0.01019	-0.1073
FC	HGB	CogTotalComp_Unadj	0.2536	0.2596	0.04469	-0.101
FC	KR	CogCrystalComp_Unadj	0.3377	0.3051	-3.094	0.06809
FC	KR	CogFluidComp_Unadj	0.1781	0.1507	-2.125	-0.01158
FC	KR	CogTotalComp_Unadj	0.3375	0.2912	-1.51	0.05574
FC	linear_BR	CogCrystalComp_Unadj	0.4508	0.44	0.1961	0.1046
FC	linear_BR	CogFluidComp_Unadj	0.3423	0.2989	0.1079	0.03286
FC	linear_BR	CogTotalComp_Unadj	0.4508	0.4235	0.1943	0.07845
SC	HGB	CogCrystalComp_Unadj	0.1722	0.162	0.01034	-0.1941
SC	HGB	CogFluidComp_Unadj	0.03564	0.01659	-0.006568	-0.1896
SC	HGB	CogTotalComp_Unadj	0.1255	0.1219	0.01152	-0.2291
SC	KR	CogCrystalComp_Unadj	0.241	0.2229	-2.468	-0.02867
SC	KR	CogFluidComp_Unadj	0.1135	0.1124	-1.603	-0.07622

Showing first 14 of 54 rows. Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n1_cognition_summary.csv.

3.0.12 Nonlinear reconstruction summary

rep	estimator	metric	median_FC_to_X	median_X_to_FC	asym_FCwins
SC	linear_PLS	demeaned_pearson	0.1355	0.0847	1.621
SC	linear_PLS	pearson	0.9125	0.8292	1.101

rep	estimator	metric	median_FC_to_X	median_X_to_FC	asym_FCwins
SC	linear_PLS	top1_acc	0.1179	0.05128	2.562
SC	linear_PLS	avg_rank	0.8787	0.7128	1.225
SC	linear_PLS	mse	0.005533	0.01429	2.594
SC	linear_PLS	r2	-0.02205	-0.05716	0.03595
SC	KR	demeaned_pearson	0.1334	0.08116	1.638
SC	KR	pearson	0.915	0.8355	1.096
SC	KR	top1_acc	0.159	0.04872	3.422
SC	KR	avg_rank	0.9104	0.7313	1.251
SC	KR	mse	0.005401	0.0138	2.578
SC	KR	r2	0.002031	-0.02052	0.02313
r2t	linear_PLS	demeaned_pearson	0.1113	0.04925	2.264
r2t	linear_PLS	pearson	0.8453	0.8153	1.037

Showing first 14 of 36 rows. Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n2_reconstruction_summary.csv.

3.0.13 Nonlinear marginal summary

estimator	metric	median_SC	median_SC_r2t	median_delta	wilcoxon_p_greater
linear	demeaned_pearson	0.0847	0.07908	-0.001289	0.9814
linear	pearson	0.8292	0.8272	-0.00138	0.999
linear	top1_acc	0.05128	0.0359	-0.01538	0.9404
linear	avg_rank	0.7128	0.702	-0.000618	0.8125
linear	mse	0.01429	0.01447	0.0001069	0.00293
linear	r2	-0.05716	-0.0638	-0.007281	0.999
KR	demeaned_pearson	0.08116	0.0735	-0.004938	0.998
KR	pearson	0.8355	0.835	-0.0001581	0.9346
KR	top1_acc	0.04872	0.04615	-0.005102	0.9492
KR	avg_rank	0.7313	0.7021	-0.02004	1
KR	mse	0.0138	0.01384	5.78e-06	0.2783
KR	r2	-0.02052	-0.02028	0.0001494	0.3848

Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n3_marginal_summary.csv.

3.0.14 Residual cognition summary

rep	target	variant	pearson	spearman	r2
FC	CogCrystalComp_Unadj	final	0.4538	0.4285	0.1975
FC	CogCrystalComp_Unadj	template	0.4516	0.4306	0.1965
FC	CogFluidComp_Unadj	final	0.3096	0.2926	0.078
FC	CogFluidComp_Unadj	template	0.3057	0.2881	0.08668
FC	CogTotalComp_Unadj	final	0.4349	0.3978	0.1847
FC	CogTotalComp_Unadj	template	0.4364	0.403	0.1723
SC	CogCrystalComp_Unadj	final	0.1923	0.1797	-0.006415
SC	CogCrystalComp_Unadj	template	0.1936	0.1803	0.02854
SC	CogFluidComp_Unadj	final	0.1201	0.1167	-0.01965
SC	CogFluidComp_Unadj	template	0.1277	0.1193	0.01068
SC	CogTotalComp_Unadj	final	0.192	0.1808	-0.00433
SC	CogTotalComp_Unadj	template	0.1972	0.1814	0.02963
SC_r2t	CogCrystalComp_Unadj	final	0.1621	0.162	-0.09939
SC_r2t	CogCrystalComp_Unadj	template	0.1646	0.1632	0.004629

Showing first 14 of 36 rows. Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n4_cog_summary.csv.

3.0.15 Residual reconstruction summary

rep	direction	metric	median_template	median_final	median_improvement	wilcoxon_p_improve
SC	FC->SC	demeaned_pearson	0.1355	0.1405	0.005203	0.0009766
SC	FC->SC	pearson	0.9125	0.9079	-0.004731	1
SC	FC->SC	top1_acc	0.1179	0.1282	0.007692	0.1797
SC	FC->SC	avg_rank	0.8787	0.8771	-0.003734	0.9473

rep	direction	metric	median_template	median_final	median_improvement	wilcoxon_p_improve
SC	FC->SC	mse	0.005533	0.005827	-0.0002896	1
SC	FC->SC	r2	-0.02205	-0.0722	-0.05126	1
SC	SC->FC	demeaned_pearson	0.0847	0.08665	0.002296	0.0009766
SC	SC->FC	pearson	0.8292	0.8179	-0.01112	1
SC	SC->FC	top1_acc	0.05128	0.04103	-0.005128	0.6963
SC	SC->FC	avg_rank	0.7128	0.7118	-0.002972	0.6523
SC	SC->FC	mse	0.01429	0.01517	-0.0009521	1
SC	SC->FC	r2	-0.05716	-0.1255	-0.0692	1
r2t	FC->r2t	demeaned_pearson	0.1113	0.1169	0.005779	0.0009766
r2t	FC->r2t	pearson	0.8453	0.834	-0.01054	1

Showing first 14 of 36 rows. Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n4_recon_summary.csv.

3.0.16 Sink cognition summary

rep	target	pearson	spearman	r2
FC	CogCrystalComp_Unadj	0.4516	0.4306	0.1965
FC	CogFluidComp_Unadj	0.3057	0.2881	0.08668
FC	CogTotalComp_Unadj	0.4364	0.403	0.1723
bv+demo	CogCrystalComp_Unadj	0.3486	0.3493	0.09952
bv+demo	CogFluidComp_Unadj	0.2978	0.2878	0.08178
bv+demo	CogTotalComp_Unadj	0.3731	0.3449	0.1322
sink_linear	CogCrystalComp_Unadj	0.4137	0.4024	0.1655
sink_linear	CogFluidComp_Unadj	0.3013	0.2957	0.08772
sink_linear	CogTotalComp_Unadj	0.4012	0.3809	0.1525
sink_residual	CogCrystalComp_Unadj	0.4147	0.3986	0.1441
sink_residual	CogFluidComp_Unadj	0.2855	0.2799	0.05765
sink_residual	CogTotalComp_Unadj	0.3809	0.363	0.1133

Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n5_cog_summary.csv.

3.0.17 Sink reconstruction summary

rep	demeaned_pearson	pearson	top1_acc	avg_rank	mse	r2
SC_linear_ref	0.0847	0.8292	0.05128	0.7128	0.01429	-0.05716
sink_linear	0.09416	0.8288	0.04872	0.7229	0.01431	-0.05738
sink_residual	0.09589	0.8179	0.05114	0.7257	0.01518	-0.1191

Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n5_recon_summary.csv.

3.0.18 Scaling summary

task	n_sub	n_seeds	median_linear	median_final	median_gap	gap_min	gap_max	wilcoxon_p_gap_gt0
cognition	100	10	0.3021	0.29	-0.008681	-0.0352	-0.00161	1
cognition	200	10	0.3267	0.3243	-0.01516	-0.032	0.0119	0.9932
cognition	400	10	0.3936	0.3797	-0.01192	-0.03402	0.0003692	0.999
cognition	682	1	0.4825	0.4861	0.003576	0.003576	0.003576	0.5
cognition	683	9	0.3914	0.3746	-0.01281	-0.03	0.005056	0.9902
reconstruction	100	10	0.04122	0.03526	-0.005361	-0.007601	0.0002357	0.999
reconstruction	200	10	0.06697	0.06547	-0.001893	-0.00573	0.005098	0.8838
reconstruction	400	10	0.09404	0.09313	-0.001541	-0.004092	0.0004269	0.9902
reconstruction	682	1	0.1098	0.1072	-0.002532	-0.002532	-0.002532	1
reconstruction	683	9	0.1098	0.1097	-0.0008874	-0.003509	0.001176	0.9805

Full source: notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n6_scaling_summary.csv.

3.0.19 Family AUC

parcellation	variant	relation	n_pairs	auc	auc_lo	auc_hi	p_perm	p_fdr	sig_fdr
Glasser	obs_SC	MZ	242	0.9989	0.9981	0.9996	0	0	True
Glasser	obs_SC	DZ	127	0.938	0.9235	0.9512	0	0	True
Glasser	obs_SC	sibling	1254	0.863	0.8499	0.8758	0	0	True
Glasser	obs_FC	MZ	242	0.9897	0.9844	0.9938	0	0	True
Glasser	obs_FC	DZ	127	0.8859	0.8523	0.9137	0	0	True
Glasser	obs_FC	sibling	1254	0.8229	0.8065	0.8385	0	0	True
Glasser	pred_SC_raw	MZ	242	0.9735	0.9647	0.9813	0	0	True
Glasser	pred_SC_raw	DZ	127	0.7982	0.7541	0.8396	0	0	True
Glasser	pred_SC_raw	sibling	1254	0.6797	0.6597	0.7002	0	0	True
Glasser	pred_SC_resid_bvdemo	MZ	242	0.9885	0.982	0.9938	0	0	True
Glasser	pred_SC_resid_bvdemo	DZ	127	0.8227	0.784	0.8604	0	0	True
Glasser	pred_SC_resid_bvdemo	sibling	1254	0.8097	0.7947	0.825	0	0	True
Glasser	combined_pred_SC	MZ	242	0.8971	0.8794	0.9145	0	0	True
Glasser	combined_pred_SC	DZ	127	0.7006	0.6543	0.742	0	0	True
Glasser	combined_pred_SC	sibling	1254	0.5049	0.482	0.5258	0.6482	0.6482	False
Glasser	bvdemo_to_SC	MZ	242	0.9507	0.9353	0.9638	0	0	True
Glasser	bvdemo_to_SC	DZ	127	0.7997	0.7591	0.8399	0	0	True
Glasser	bvdemo_to_SC	sibling	1254	0.5628	0.541	0.5851	0	0	True
Glasser	pred_FC_raw	MZ	242	0.9104	0.8885	0.9293	0	0	True
Glasser	pred_FC_raw	DZ	127	0.7557	0.7152	0.7923	0	0	True
Glasser	pred_FC_raw	sibling	1254	0.7097	0.6907	0.7297	0	0	True
Glasser	pred_FC_resid_bvdemo	MZ	242	0.9181	0.8964	0.9372	0	0	True
Glasser	pred_FC_resid_bvdemo	DZ	127	0.7413	0.6924	0.785	0	0	True
Glasser	pred_FC_resid_bvdemo	sibling	1254	0.7429	0.7254	0.7607	0	0	True

Showing first 24 of 48 rows. Full source: reproduction/ family_mechanism/ outputs/ family_auc.csv.

3.0.20 F8 stability

parcellation	anchor_pc	median_abs_cos	min_abs_cos	median_expl_var	median_FC_to_PC_R2	median_AUC_MZ	median_AUC_DZ	median_AUC_
Glasser	1	0.9849	0.9812	0.03826	0.5921	0.8059	0.6194	0.4393
Glasser	2	0.9345	0.8329	0.01593	0.0169	0.64	0.4705	0.5662
Glasser	3	0.8856	0.8253	0.01496	0.2247	0.7108	0.5993	0.581
Glasser	4	0.8956	0.8483	0.01347	0.1204	0.6805	0.5994	0.5796
Glasser	5	0.8675	0.7553	0.01185	0.1098	0.6061	0.5397	0.5629
Glasser	6	0.7835	0.6339	0.01079	0.06598	0.7189	0.6338	0.5633
Glasser	7	0.7122	0.5642	0.01016	0.09105	0.671	0.5914	0.5591
Glasser	8	0.7558	0.6114	0.009241	0.03474	0.5946	0.5613	0.543
Glasser	9	0.6278	0.4594	0.008823	0.03123	0.6229	0.5516	0.5491
Glasser	10	0.5658	0.5056	0.008766	0.06259	0.6149	0.5454	0.5387
4S456Parcels	1	0.983	0.9794	0.03828	0.5767	0.8132	0.6354	0.4438
4S456Parcels	2	0.9662	0.9445	0.01901	0.07535	0.7326	0.5198	0.5534
4S456Parcels	3	0.9418	0.8673	0.0146	0.02866	0.6587	0.5076	0.5571
4S456Parcels	4	0.9141	0.838	0.01355	0.2361	0.7214	0.5987	0.5925
4S456Parcels	5	0.8577	0.7175	0.01225	0.124	0.6316	0.5893	0.5477
4S456Parcels	6	0.8454	0.7143	0.01136	0.07854	0.6913	0.6384	0.5654
4S456Parcels	7	0.8431	0.6393	0.01045	0.1307	0.6856	0.5616	0.5779
4S456Parcels	8	0.7662	0.4949	0.009417	0.07259	0.6773	0.5627	0.5385
4S456Parcels	9	0.6768	0.4602	0.008707	0.144	0.7218	0.58	0.5759
4S456Parcels	10	0.4513	0.2749	0.008636	0.0991	0.6476	0.5424	0.5481

Full source: reproduction/ family_mechanism/ outputs/ f8_stability.csv.

4 Existing-Data Gap Closure

This section answers the review-style holes that can be closed from files already present in the repository. It does not claim to solve checks that require raw diffusion repeat data, new objectives, or additional cohorts; those are triaged in the next section.

4.1 E1. Reconstruction Quality Does Not Track Cognitive Lift

parcellation	reconstruction_metric	n	pearson_r_with_lift	pearson_p	spearman_r_with_lift	spearman_p
4S456Parcels	demeaned_pearson	330	-0.070	0.204	-0.252	3.71e-06
4S456Parcels	avg_rank	330	0.148	0.007	0.139	0.011
4S456Parcels	top1_acc	330	-0.058	0.289	0.007	0.905
Glasser	demeaned_pearson	330	-0.066	0.233	-0.148	0.007

parcellation	reconstruction_metric	n	pearson_r_with_lift	pearson_p	spearman_r_with_lift	spearman_p
Glasser	avg_rank	330	0.043	0.439	0.043	0.433
Glasser	top1_acc	330	0.131	0.017	0.148	0.007

Interpretation: the strongest reconstruction scores are not the cells that deliver cognitive gain. Across 660 joined rows, the FC->SC reconstruction metrics are near orthogonal to lift over the brain-volume/demographic baseline. This directly closes the hole that the null might be a presentation artifact of looking at the wrong reconstruction metric.

Source: joined: reproduction/ outputs/ reconstruction.csv; reproduction/ outputs/ downstream.csv.

4.2 E2. Existing Target Sweep: What the Current Grid Already Covers

target	input_set	metric	mean_metric	mean_lift	median_perm_p	n_cells
CogCryst	obs_FC	pearson	0.482	0.128	0.039	20
CogCryst	obs_FC+bv+demo	pearson	0.508	0.154	0.004	20
CogCryst	obs_SC	pearson	0.256	-0.098	0.869	20
CogCryst	pred_SC	pearson	0.343	-0.011	0.582	20
CogCryst	pred_SC+bv+demo	pearson	0.412	0.058	0.037	20
CogFluid	obs_FC	pearson	0.324	0.041	0.333	20
CogFluid	obs_FC+bv+demo	pearson	0.351	0.068	0.138	20
CogFluid	obs_SC	pearson	0.164	-0.119	0.974	20
CogFluid	pred_SC	pearson	0.252	-0.031	0.861	20
CogFluid	pred_SC+bv+demo	pearson	0.315	0.032	0.249	20
CogTotal	obs_FC	pearson	0.446	0.087	0.084	20
CogTotal	obs_FC+bv+demo	pearson	0.471	0.112	0.019	20
CogTotal	obs_SC	pearson	0.252	-0.107	0.930	20
CogTotal	pred_SC	pearson	0.348	-0.011	0.651	20
CogTotal	pred_SC+bv+demo	pearson	0.403	0.044	0.123	20
age	obs_FC	pearson	0.449	-0.387	0.500	20
age	obs_FC+bv+demo	pearson	0.788	-0.049	5.00e-04	20
age	obs_SC	pearson	0.360	-0.476	0.500	20
age	pred_SC	pearson	0.296	-0.540	0.500	20
age	pred_SC+bv+demo	pearson	0.996	0.159	5.00e-04	20
sex	obs_FC	balanced_acc	0.926	-0.010		20
sex	obs_FC+bv+demo	balanced_acc	0.993	0.056		20
sex	obs_SC	balanced_acc	0.919	-0.018		20
sex	pred_SC	balanced_acc	0.887	-0.050		20
sex	pred_SC+bv+demo	balanced_acc	1.000	0.063		20

Interpretation: the current target sweep is not broad enough to claim all behavior, but it already covers three cognition composites plus age and sex controls. Observed FC plus bv+demo reliably improves cognition; predicted SC alone does not. Predicted SC plus bv+demo gives a small, target-dependent lift, strongest for crystallized cognition and age-like information. Sex is best read as a leakage/control target because bv+demo nearly solves it.

Source: Bayesian-ridge downstream grid: reproduction/ outputs/ downstream.csv.

4.3 E3. Seed-Level Lift Bounds For the Current Cognition Null

parcellation	input_set	n_seed_target_cells	mean_lift	ci95_lo	ci95_hi	ci_halfwidth	frac_perm_p_lt_0.05	median_perm_p
4S456Parcels	pred_SC	30	-0.009	-0.042	0.024	0.033	0.067	0.712
Glasser	pred_SC	30	-0.027	-0.051	-0.003	0.024	0.000	0.711
4S456Parcels	pred_SC+bv+demo	30	0.049	0.028	0.069	0.021	0.400	0.161
Glasser	pred_SC+bv+demo	30	0.041	0.027	0.054	0.014	0.333	0.130
4S456Parcels	obs_FC+bv+demo	30	0.106	0.080	0.131	0.025	0.600	0.039
Glasser	obs_FC+bv+demo	30	0.117	0.090	0.143	0.026	0.633	0.028

Interpretation: with the existing ten frozen splits and three cognition targets, predicted SC alone is bounded near zero or below. Predicted SC plus bv+demo has a small positive lift, but it is well below observed FC plus bv+demo. This makes the null more quantitative: the current grid can detect an observed-FC-size improvement, but predicted SC is not delivering one.

Source: seed-level downstream lift: reproduction/ outputs/ downstream.csv.

4.4 E4. Leak Audit Summary

verdict	n
ok	3344
EXEMPT_FLAGGED	1052
EXPECTED_SIGNAL	4

By verdict and target:

verdict	target	n
EXEMPT_FLAGGED	age	416
EXEMPT_FLAGGED	sex	636
EXPECTED_SIGNAL	sex	4
ok	age	1784
ok	sex	1560

Interpretation: the grid has zero genuine LEAK_FAIL cells. The only threshold exceedances are either expected raw-connectome sex/age signal or cells explicitly marked as containing bv+demo.

Source: leak verdict: reproduction/ outputs/ leak_verdict.csv.

4.5 E5. Objective-Mismatch Evidence Already Present

parcellation	variant	auc	auc_lo	auc_hi	p_fdr	sig_fdr
4S456Parcels	bvdemo_to_SC	0.565	0.544	0.586	0.000	yes
4S456Parcels	combined_pred_SC	0.501	0.479	0.523	0.904	no
4S456Parcels	obs_FC	0.821	0.805	0.836	0.000	yes
4S456Parcels	obs_SC	0.884	0.873	0.896	0.000	yes
4S456Parcels	pred_FC_raw	0.726	0.708	0.745	0.000	yes
4S456Parcels	pred_FC_resid_bvdemo	0.769	0.753	0.785	0.000	yes
4S456Parcels	pred_SC_raw	0.670	0.650	0.690	0.000	yes
4S456Parcels	pred_SC_resid_bvdemo	0.816	0.801	0.831	0.000	yes
Glasser	bvdemo_to_SC	0.563	0.541	0.585	0.000	yes
Glasser	combined_pred_SC	0.505	0.482	0.526	0.648	no
Glasser	obs_FC	0.823	0.807	0.839	0.000	yes
Glasser	obs_SC	0.863	0.850	0.876	0.000	yes
Glasser	pred_FC_raw	0.710	0.691	0.730	0.000	yes
Glasser	pred_FC_resid_bvdemo	0.743	0.725	0.761	0.000	yes
Glasser	pred_SC_raw	0.680	0.660	0.700	0.000	yes
Glasser	pred_SC_resid_bvdemo	0.810	0.795	0.825	0.000	yes

Interpretation: predicted connectomes can preserve family/identity signal when the objective or representation selects for it. The same artifact does not automatically translate into cognition gain. This strengthens the paper's objective-mismatch claim: the model is not universally incapable; it is carrying the wrong signal for the downstream cognition task.

Source: family mechanism: reproduction/ family_mechanism/ outputs/ family_auc.csv.

4.6 E6. Per-Subject FC Reliability Is Not Driving the SC->FC Result

source	n	pearson_achieved_vs_ceiling	pearson_p	spearman_achieved_vs_ceiling	spearman_p
SC	857	0.012	0.722	-9.67e-04	0.977
bv+demo	857	0.147	1.66e-05	0.147	1.66e-05

Interpretation: the SC achieved-vs-ceiling correlation is essentially flat, while the bv+demo baseline detects a reliability association. This is an internal positive control: the analysis can see reliability structure when it exists, but SC prediction is not explained by per-subject FC reliability.

Source: per-subject reliability: notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ h_correlations.csv; notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ h_per_subject_achieved_vs_ceiling.csv.

5 New Data and Experiment Triage

The existing data close several presentation-level holes, but not every scientific hole. The remaining checks below require new model runs, saved prediction matrices, raw diffusion perturbations, or an external cohort. A standalone copy of this triage is written to `sanity_check_gap_plan.md`.

6 Sanity-Check Gap Plan

This file separates checks already closed from existing repository outputs from checks that require new data, saved predictions, or new model runs.

6.1 Closed With Existing Data

- Reconstruction quality is decoupled from cognitive lift; see `supplement_sanity_checks.md`, section Existing-Data Gap Closure.
- The current downstream grid covers three cognition composites plus age/sex controls.
- Seed-level lift bounds show predicted SC alone is near-zero or negative for cognition, while observed FC plus bv+demo is detectably positive.
- Leak verdicts show zero genuine LEAK_FAIL cells.
- Family AUC shows predicted connectomes can preserve identity/family signal when the representation selects for it.
- Per-subject FC reliability does not explain SC->FC achieved performance.

6.2 Requires New Data Or New Runs

- **SC-side reliability/noise**
 - Needed: Repeat dMRI, split-half tractography, bootstrap streamlines, or saved tractography perturbations.
 - Feasibility: medium if raw diffusion/tractography pipeline is available; low from summary CSVs alone.
 - Why it matters: Current repository bounds FC noise well but cannot produce a true SC test-retest ceiling.
- **Objective interpolation**
 - Needed: New training grid mixing reconstruction, fingerprint/family, and cognition objectives on frozen splits.
 - Feasibility: medium-high computationally; no new cohort needed.
 - Why it matters: Would turn the objective-mismatch argument from inferential to causal.
- **Broader phenotype families**
 - Needed: Additional HCP behavioral, personality, motor, emotion, and latent phenotype targets.
 - Feasibility: high if phenotypes are already local; medium if target cleaning is needed.
 - Why it matters: Current grid covers three cognition composites plus age/sex controls, not all behavior.
- **Predicted-SC calibration/topology**
 - Needed: Saved predicted matrices or regenerated predictions; compare degree, strength, sparsity, modularity, hubs.
 - Feasibility: medium if predictions are cached; medium-low if all predictions must be regenerated.
 - Why it matters: Correlation can look acceptable while graph topology is biologically distorted.
- **Edge-class stratification**
 - Needed: Per-edge predictions plus distance, network labels, reliability, and streamline-strength bins.
 - Feasibility: medium with saved matrices and atlas metadata.
 - Why it matters: Could show whether useful signal is concentrated in short/long, intra/inter-network, or high-reliability edges.

- **Split stress tests**
 - Needed: Rerun selected cells under random, family-aware, age/sex-balanced, high-motion-excluded, and low-motion-only splits.
 - Feasibility: medium; compute-heavy but no new data.
 - Why it matters: Would make the split-drift/leakage defense harder to attack.
- **External validity**
 - Needed: Second cohort with FC, dMRI-derived SC, demographics, and comparable cognition/behavior targets.
 - Feasibility: low-medium depending on access.
 - Why it matters: HCP-YA-only evidence supports an internal claim, not a universal population claim.
- **Full hyperparameter leakage audit**
 - Needed: Static/code audit plus small rerun proving PCA, scaling, residualization, and target transforms fit inside folds.
 - Feasibility: high for code audit; medium for rerun.
 - Why it matters: Leak verdicts cover output behavior; this would document every transform boundary.

7 Detailed Findings Notes

The following subsections preserve the detailed findings notes that motivated and defended the manuscript claims. Minor Unicode normalization is applied for stable LaTeX compilation; source paths are listed in the manifest.

7.1 Reproduction Grid Findings

7.1.1 Reproduction Grid -- Findings

Generated by summarize.py. Metric reporting order: reconstruction leads with demeaned_r/avg_rank; downstream with lift_over_bvdemo + paired-permutation p. **Inspect the 4S456 F1-F5 cells first** (the genuinely-new cross-parcellation evidence).

Reconstruction (demeaned_pearson, mean+/-std over seeds)

4S456Parcels (estimator=pca_pls)

input -> target	demeaned_r	avg_rank	top1
FC -> SC	0.146+/-0.003	0.897+/-0.010	0.151+/-0.017
SC -> FC	0.087+/-0.006	0.730+/-0.017	0.049+/-0.013
bv -> SC	0.185+/-0.005	0.903+/-0.006	0.129+/-0.012
bv -> FC	0.045+/-0.005	0.625+/-0.013	0.010+/-0.009
demo -> SC	0.133+/-0.009	0.764+/-0.028	0.023+/-0.008
demo -> FC	0.105+/-0.009	0.677+/-0.020	0.013+/-0.010
bv+demo -> SC	0.191+/-0.007	0.908+/-0.011	0.173+/-0.024
bv+demo -> FC	0.094+/-0.006	0.687+/-0.017	0.024+/-0.013
FC+bv+demo -> SC	0.184+/-0.005	0.938+/-0.009	0.232+/-0.016
SC+bv+demo -> FC	0.106+/-0.007	0.758+/-0.018	0.055+/-0.018
FC -> FC	0.418+/-0.034	1.000+/-0.000	0.997+/-0.004
SC -> SC	0.386+/-0.030	1.000+/-0.000	1.000+/-0.000

Asymmetry FC->SC / SC->FC = 1.68x (FC->SC=0.146, SC->FC=0.087).

Ceiling B (FC->FC oracle): bayesian_ridge=0.632, kernel_ridge=0.631, pca_pls=0.418

Glasser (estimator=pca_pls)

input -> target	demeaned_r	avg_rank	top1
FC -> SC	0.136+/-0.007	0.875+/-0.015	0.120+/-0.022
SC -> FC	0.084+/-0.008	0.714+/-0.019	0.049+/-0.011
bv -> SC	0.162+/-0.007	0.864+/-0.013	0.086+/-0.014

input -> target	demeaned_r	avg_rank	top1
bv -> FC	0.049+/-0.005	0.623+/-0.009	0.012+/-0.008
demo -> SC	0.130+/-0.009	0.754+/-0.027	0.023+/-0.014
demo -> FC	0.111+/-0.010	0.671+/-0.021	0.013+/-0.006
bv+demo -> SC	0.169+/-0.009	0.875+/-0.015	0.129+/-0.015
bv+demo -> FC	0.100+/-0.008	0.685+/-0.019	0.024+/-0.006
FC+bv+demo -> SC	0.171+/-0.006	0.914+/-0.014	0.169+/-0.019
SC+bv+demo -> FC	0.105+/-0.007	0.743+/-0.022	0.054+/-0.015
FC -> FC	0.451+/-0.026	1.000+/-0.000	0.999+/-0.002
SC -> SC	0.376+/-0.018	1.000+/-0.000	1.000+/-0.000

Asymmetry FC->SC / SC->FC = 1.61x (FC->SC=0.136, SC->FC=0.084).

Ceiling B (FC->FC oracle): bayesian_ridge=0.673, kernel_ridge=0.672, pca_pls=0.451

Downstream cognition (lift over bv+demo, mean+/-std; median paired-perm p)

4S456Parcels (estimator=bayesian_ridge)

input	target	pearson	lift_over_bvdemo	median perm p	residualized_r
bv+demo	CogTotal	0.359+/-0.086	0.000+/-0.000	1	0.022+/-0.072
bv+demo	CogFluid	0.283+/-0.072	0.000+/-0.000	1	-0.003+/-0.111
bv+demo	CogCryst	0.354+/-0.060	0.000+/-0.000	1	0.054+/-0.091
obs_FC	CogTotal	0.445+/-0.073	0.086+/-0.055	0.0905	0.289+/-0.076
obs_FC	CogFluid	0.321+/-0.079	0.037+/-0.035	0.347	0.204+/-0.072
obs_FC	CogCryst	0.476+/-0.072	0.122+/-0.087	0.0407	0.352+/-0.087
obs_SC	CogTotal	0.256+/-0.076	-0.103+/-0.055	0.927	0.057+/-0.067
obs_SC	CogFluid	0.165+/-0.108	-0.118+/-0.055	0.971	0.030+/-0.068
obs_SC	CogCryst	0.259+/-0.065	-0.095+/-0.077	0.883	0.066+/-0.084
obs_FC+obs_SC	CogTotal	0.446+/-0.085	0.087+/-0.053	0.073	0.283+/-0.079
obs_FC+obs_SC	CogFluid	0.326+/-0.102	0.043+/-0.047	0.228	0.205+/-0.083
obs_FC+obs_SC	CogCryst	0.475+/-0.083	0.121+/-0.091	0.0335	0.339+/-0.097
pred_SC	CogTotal	0.354+/-0.100	-0.005+/-0.085	0.612	0.182+/-0.106
pred_SC	CogFluid	0.273+/-0.078	-0.010+/-0.062	0.843	0.152+/-0.064
pred_SC	CogCryst	0.342+/-0.105	-0.012+/-0.117	0.629	0.184+/-0.126
pred_FC	CogTotal	0.271+/-0.068	-0.088+/-0.084	0.871	0.079+/-0.087
pred_FC	CogFluid	0.211+/-0.100	-0.072+/-0.070	0.906	0.091+/-0.079
pred_FC	CogCryst	0.245+/-0.062	-0.109+/-0.087	0.896	0.025+/-0.089
obs_FC+bv+demo	CogTotal	0.468+/-0.078	0.109+/-0.047	0.0192	0.292+/-0.077
obs_FC+bv+demo	CogFluid	0.345+/-0.092	0.062+/-0.043	0.158	0.208+/-0.075
obs_FC+bv+demo	CogCryst	0.500+/-0.072	0.146+/-0.081	0.008	0.353+/-0.092
obs_SC+bv+demo	CogTotal	0.308+/-0.083	-0.052+/-0.049	0.831	0.057+/-0.076
obs_SC+bv+demo	CogFluid	0.206+/-0.115	-0.077+/-0.057	0.934	0.027+/-0.074
obs_SC+bv+demo	CogCryst	0.312+/-0.056	-0.042+/-0.059	0.695	0.068+/-0.091
pred_SC+bv+demo	CogTotal	0.407+/-0.100	0.048+/-0.055	0.143	0.179+/-0.111
pred_SC+bv+demo	CogFluid	0.325+/-0.080	0.042+/-0.043	0.216	0.146+/-0.062
pred_SC+bv+demo	CogCryst	0.410+/-0.081	0.056+/-0.069	0.0477	0.186+/-0.132
pred_FC+bv+demo	CogTotal	0.334+/-0.067	-0.025+/-0.061	0.735	0.082+/-0.084
pred_FC+bv+demo	CogFluid	0.260+/-0.099	-0.024+/-0.058	0.765	0.090+/-0.079
pred_FC+bv+demo	CogCryst	0.326+/-0.052	-0.028+/-0.057	0.542	0.028+/-0.085

Glasser (estimator=bayesian_ridge)

input	target	pearson	lift_over_bvdemo	median perm p	residualized_r
bv+demo	CogTotal	0.359+/-0.086	0.000+/-0.000	1	0.022+/-0.072
bv+demo	CogFluid	0.283+/-0.072	0.000+/-0.000	1	-0.003+/-0.111
bv+demo	CogCryst	0.354+/-0.060	0.000+/-0.000	1	0.054+/-0.091
obs_FC	CogTotal	0.447+/-0.082	0.088+/-0.065	0.0802	0.297+/-0.088
obs_FC	CogFluid	0.327+/-0.082	0.044+/-0.044	0.326	0.216+/-0.075
obs_FC	CogCryst	0.487+/-0.070	0.133+/-0.093	0.0307	0.378+/-0.092
obs_SC	CogTotal	0.248+/-0.080	-0.111+/-0.065	0.939	0.041+/-0.081
obs_SC	CogFluid	0.163+/-0.102	-0.120+/-0.045	0.977	0.008+/-0.061
obs_SC	CogCryst	0.253+/-0.068	-0.101+/-0.086	0.817	0.061+/-0.089
obs_FC+obs_SC	CogTotal	0.449+/-0.086	0.090+/-0.065	0.075	0.289+/-0.090
obs_FC+obs_SC	CogFluid	0.331+/-0.098	0.048+/-0.047	0.212	0.207+/-0.079
obs_FC+obs_SC	CogCryst	0.490+/-0.085	0.136+/-0.100	0.022	0.373+/-0.109
pred_SC	CogTotal	0.341+/-0.072	-0.018+/-0.055	0.651	0.166+/-0.052
pred_SC	CogFluid	0.230+/-0.085	-0.053+/-0.062	0.875	0.117+/-0.077
pred_SC	CogCryst	0.344+/-0.072	-0.010+/-0.072	0.582	0.195+/-0.056
pred_FC	CogTotal	0.247+/-0.075	-0.113+/-0.052	0.94	0.043+/-0.084
pred_FC	CogFluid	0.186+/-0.092	-0.097+/-0.045	0.953	0.051+/-0.066

input	target	pearson	lift_over_bvdemo	median perm p	residualized_r
pred_FC	CogCryst	0.219+/-0.078	-0.135+/-0.072	0.96	-0.000+/-0.112
obs_FC+bv+demo	CogTotal	0.474+/-0.082	0.115+/-0.055	0.0157	0.303+/-0.085
obs_FC+bv+demo	CogFluid	0.356+/-0.093	0.073+/-0.044	0.0952	0.219+/-0.074
obs_FC+bv+demo	CogCryst	0.516+/-0.066	0.162+/-0.081	0.00225	0.382+/-0.092
obs_SC+bv+demo	CogTotal	0.309+/-0.091	-0.050+/-0.053	0.824	0.043+/-0.089
obs_SC+bv+demo	CogFluid	0.217+/-0.103	-0.066+/-0.043	0.937	0.010+/-0.062
obs_SC+bv+demo	CogCryst	0.319+/-0.059	-0.035+/-0.065	0.604	0.061+/-0.094
pred_SC+bv+demo	CogTotal	0.399+/-0.086	0.040+/-0.027	0.123	0.160+/-0.045
pred_SC+bv+demo	CogFluid	0.305+/-0.084	0.022+/-0.038	0.51	0.111+/-0.091
pred_SC+bv+demo	CogCryst	0.415+/-0.071	0.061+/-0.035	0.037	0.191+/-0.062
pred_FC+bv+demo	CogTotal	0.321+/-0.078	-0.038+/-0.042	0.832	0.048+/-0.077
pred_FC+bv+demo	CogFluid	0.245+/-0.096	-0.038+/-0.043	0.866	0.051+/-0.065
pred_FC+bv+demo	CogCryst	0.314+/-0.069	-0.040+/-0.044	0.813	0.004+/-0.108

Leak checks

- LEAK_FAIL (genuine: demographic-free, non-connectome input over threshold): **0**
- EXPECTED_SIGNAL (raw connectomes predict sex/age -- real biology, not a leak): **4**
- EXEMPT_FLAGGED (contain bv+demo; cognition-only): **1052**
- ok: **3344**

7.2 Grid Exploration Audit

7.2.1 Grid Exploration -- Do the Findings Hold, and What Else Is in There?

Deep double-check of the 13,640-cell reproduction grid, **straight from the merged CSVs** (outputs/{reconstruction,downstream,leak_verdict}.csv). Regenerate with `python exploration/explore.py (-> digest.txt)` and `python exploration/make_figures.py (-> figures/)`. Everything below is computed, not remembered.

Bottom line: every headline claim (F1, F2, F4, F5, Ceiling B) replicates on *both* parcellations and is stable across seeds. The interesting part is in the margins -- eight things the summary report doesn't say, three of which actually *strengthen* the story and two of which are methodological flags. Verdicts at the end.

Part 1 -- Do the confirmatory findings line up? (Yes.)

Claim	Result (both parcs)	Status
F1 asymmetry FC->SC > SC->FC	1.68x (4S456), 1.61x (Glasser), pca_pls	[pass] replicates
F2 dissociation anatomy vs demographics	clean double dissociation (see below)	[pass] replicates, stronger than stated
F4 FC->cognition real over baseline	obs_FC lifts CogCryst 60% of seeds, p~0.03-0.04	[pass] replicates
F5 SC underperforms / pred adds nothing	obs_SC lift < 0; pred_FC <i>actively harmful</i>	[pass] replicates, stronger
Ceiling B within-modal oracle	FC->FC 0.63/0.67, SC->SC 0.62/0.65 (BR)	[pass] replicates

All match reports/reproduction_findings.md to the third decimal. No cell is non-finite; the only NaN pearsons are the 2,200 sex rows (classification -> uses balanced_acc, by design).

Part 2 -- Eight things worth knowing (the "anything unusual" hunt)

1. The cross-modal asymmetry is NOT a reliability artifact [note] (strengthens F1) The obvious skeptic's objection to "FC->SC beats SC->FC" is "SC is just noisier, so it's a worse *target*." The oracle kills that: at the within-modal ceiling, **FC and SC are essentially equally self-predictable** -- FC->FC / SC->SC = **1.02x (4S456), 1.04x (Glasser)**. Both modalities are ~equally reliable, yet FC->SC beats SC->FC by **1.6-1.7x**. So the asymmetry is a genuine *directional information* effect (FC contains more about SC than vice-versa), not a target-noise artifact. This is a defensible, reviewer-proof framing of F1.

2. Clean double dissociation: anatomy->structure, demographics->function [note] (strengthens F2) Not just "bv predicts SC." It's a crossed dissociation, on both parcellations (figures/f2_dissociation.png):

	->SC (structure)	->FC (function)
bv (anatomy proxies)	0.185 / 0.162 (wins)	0.045 / 0.049
demo (demographics)	0.133 / 0.130	0.105 / 0.111 (wins)
ratio bv/demo	1.39 / 1.24	0.43 / 0.44

Anatomy is the better predictor *of structure*; demographics is >2x better *of function*. A clean 2x2 crossover is a much stronger statement than a single main effect.

3. pred_FC is actively harmful downstream, not merely useless [note] (sharpens F5) figures/f3_downstream_lift.png. The imputed connectomes don't just fail to help cognition -- **pred_FC drives prediction below the bv+demo baseline by -0.11 to -0.135** (the most negative cell in the whole downstream grid, both parcs). pred_SC is roughly neutral (~0). So "imputation doesn't transfer to cognition" understates it: imputed FC injects structured noise that *beats the baseline down*. obs_SC is also net-negative for cognition. Only **obs_FC** (and obs_FC+bv+demo) lifts cognition.

4. The kernel_ridge 3x3 sweep is degenerate -- 9 variants ~ 1 (compute flag) Across the entire 9-cell gammaxalpha grid, the within-seed max-min spread of demeaned_pearson is **0.004-0.005** -- i.e. the HPs do essentially nothing (digest.txt section 2). 9 of the 11 estimator-variants carry near-identical information. **Implication:** a future re-run could collapse KR to a single HP and shed ~2/3 of the estimator axis (and a big chunk of the downstream blow-up the PCA-cache note targets) with no loss of signal. Worth noting in the methods as "HP-insensitive."

5. Estimator choice moves every headline magnitude (reporting flag) The report mixes estimators: F1 cross-modal uses **pca_pls**, Ceiling B uses **bayesian_ridge**. But BR is the stronger estimator *everywhere*:

- Oracle: BR FC->FC = 0.63-0.67 vs **pca_pls only 0.42-0.45** -- PLS reaches ~65% of BR's ceiling.
- Cross-modal: BR FC->SC = 0.171/0.166 vs pca_pls 0.146/0.136 -- **BR is ~17% higher.**

The asymmetry *ratio* is estimator-robust (good), but the **absolute headline numbers depend on which estimator you quote**. Recommend stating the estimator next to every number, and consider leading F1 with BR (the best estimator) rather than PLS, for consistency with Ceiling B. Downstream is the same story inverted: **pca_pls gives ~1.5x larger lifts** than BR (obs_FC->CogCryst: pca ~ 0.18-0.20 vs BR ~ 0.12-0.13) -- but BR is the conservative choice and is correctly what the report leads with.

6. The finer parcellation specifically boosts structure prediction (structured, not noise) 4S456 vs Glasser is *not* a wash -- the difference is signed and consistent (figures/f4_cross_parcellation.png): **every ->SC pair is higher on 4S456 (+8 to +14.5%), every ->FC pair is slightly lower (-5 to -7%)**. The +60% finer parcellation adds resolvable structural detail (bv->SC +14.5%, bv+demo->SC +12.9%) but marginally dilutes functional prediction. This is the genuinely-new cross-parcellation evidence and it has a clean interpretation, not just "numbers track."

7. Cross-modal prediction captures only ~20-24% of the achievable ceiling (honest framing) FC->SC (pca_pls) / SC->SC oracle (BR) = **23.6% (4S456), 21.0% (Glasser)** (figures/f5_ceiling_gap.png). The cross-modal map is real but recovers only ~1/5-1/4 of what's in principle recoverable about the target connectome. Good honest ceiling language for the paper: "well above chance, far below the within-modal oracle."

8. Only the combined FC+SC crosses the sex-leak threshold (mild curiosity) All 4 EXPECTED_SIGNAL rows are **obs_FC+obs_SC -> sex** at score 0.991-0.995 (threshold 0.99). Neither FC nor SC alone crosses; combining the two modalities pushes sex-decodability just over the line.

Real biology (sex is strongly encoded in connectomes), correctly *not* flagged as a leak -- but a nice illustration that FC and SC carry partly *complementary* demographic information. 0 LEAK_FAIL.

Part 3 -- Smaller observations

- **CogCryst is the only reliably FC-predictable cognition.** CogCryst lifts hit 60-80% of seeds significant; CogFluid **never** reaches significance from any connectome input (0% seeds for obs_FC). Crystallized > fluid intelligence in connectome-predictability -- consistent with the literature.
 - **Identifiability ladder** (top1 fingerprint accuracy, pca_pls): FC->FC oracle ~ **0.997-0.999** (near-perfect), FC->SC cross-modal only 0.12-0.15, SC->FC worst (0.049). Adding bv+demo *raises* identifiability (FC+bv+demo->SC top1 0.17-0.23 > FC->SC alone) -- subject-info sharpens the fingerprint.
 - **Connectome adds ~nothing over bv+demo for reconstruction.** FC+bv+demo->SC vs bv+demo->SC is -0.007 (4S456) / +0.002 (Glasser): once you have anatomy+demographics, the *opposite* connectome is redundant for predicting SC. SC+bv+demo->FC adds a little more (+0.005 to +0.012). Consistent with 2..
 - **Seed stability is excellent.** Headline CVs: FC->SC just **1.8% (4S456) / 5.0% (Glasser)**; the noisiest headline is SC->FC at ~7-9%. Nothing is seed-fragile.
 - **age "prediction" from bv+demo (0.83-0.84) is near-trivial** -- age is *in* demo. The real signal is connectomes predicting age much more weakly (obs_FC ~ 0.43-0.47), and pred_* degrade it (0.25-0.32).
-

Part 4 -- Verdicts Confirmatory: [pass] All of F1, F2, F4, F5, Ceiling B replicate on both parcellations, stable across 10 seeds, no data-integrity issues. The grid is sound.

Net-new / actionable, in priority order:

1. **Use 1., 2., 3. in the paper -- they strengthen the story** (asymmetry isn't a reliability artifact; F2 is a true double dissociation; pred_FC is harmful, not just useless).
2. **Reporting hygiene (5.):** state the estimator next to every headline number; the absolute values differ by ~17% (recon) to ~1.5x (downstream) by estimator even though directions are robust.
3. **Compute (4.):** the KR 9-HP sweep is degenerate -- collapse to 1 HP on any re-run.
4. **Framing (6., 7.):** the 4S456->SC boost is a real interpretable effect; lead the ceiling discussion with "~1/5-1/4 of the within-modal oracle."

Nothing alarming surfaced -- no leak failures, no implausible cells, no seed-fragile headline, no sign flips across parcellations.

7.3 Estimator Discrepancy Resolution

7.3.1 Resolving the Two Flagged Discrepancies (oracle 0.647 vs 0.38; cognition 0.359 vs 0.129)

An analyst comparing the grid CSVs against the older PDF flagged two worrying gaps. **Both have the same root cause and neither indicates a problem with the grid:** the analyst read the `pca_pls` rows; the PDF/report numbers are `bayesian_ridge`. Hold the estimator fixed and everything matches to the third decimal. Verified empirically below (python `exploration/explore.py` reproduces it).

Discrepancy 1 -- the oracle ceiling (the one that worried you most) **Claim:** "SC->SC oracle is 0.38 here vs 0.647 in the PDF -- maybe 0.647 was raw pearson or a different oracle construction; the disattenuation denominator may be wrong."

Resolution: it's the same metric (demeaned-r), same pipeline -- just a different estimator.

SC->SC oracle (demeaned_pearson)	pca_pls	bayesian_ridge	kernel_ridge
Glasser	0.376	0.648	0.645
4S456	0.386	0.622	0.618

The PDF's **0.647 ~ bayesian_ridge SC->SC = 0.648 (Glasser)** -- a 0.001 match. The 0.38 the analyst saw is the `pca_pls` row. It is **not** raw pearson: raw pearson for SC->SC is **0.92-0.95** (and r^2 is ~0.05-0.10). So three candidate values per cell:

- demeaned-r (the PRIMARY metric): `pca_pls` 0.38 / **BR 0.648** / KR 0.645
- raw pearson (population-mean-dominated, not reportable): ~0.95
- r^2 : ~0.05-0.10

Conclusion: the ceiling did not change. The disattenuation denominator is fine -- 0.647 was the BR oracle all along. The only fix needed is reporting hygiene: the PDF quotes F1 cross-modal with `pca_pls` but the oracle with `bayesian_ridge`. **Quote one estimator throughout.** Since BR is the strongest estimator (its oracle is ~50% higher than PLS's, because PLS only reaches ~65% of BR's ceiling), the cleanest fix is to lead *everything* in BR:

BR, demeaned-r	Glasser	4S456
FC->SC (cross-modal)	0.166	0.171
SC->FC (cross-modal)	0.104	0.101
FC->FC oracle	0.673	0.632
SC->SC oracle	0.648	0.622

The asymmetry **ratio** is estimator-robust (~1.6x in both PLS and BR), so F1 is unaffected; only the absolute headline numbers move with the estimator.

Discrepancy 2 -- cognition baseline dropped (bv+demo 0.359 -> 0.129) **Claim:** "bv+demo CogTotal is 0.129 here vs 0.359 in the PDF; obs_FC CogCryst 0.439 vs 0.434. The baseline dropped a lot, which widens FC's lift, but perm-p still isn't significant, so variance must be higher -- why did bv+demo cognition drop?"

Resolution: same estimator confusion. The analyst's numbers are `pca_pls`, exactly:

Glasser, pearson	pca_pls	bayesian_ridge (= PDF)
bv+demo -> CogTotal	0.129 <- analyst	0.359 <- PDF
obs_FC -> CogCryst	0.439 <- analyst	0.487

0.129 and 0.439 are the `pca_pls` Glasser values to the digit. The baseline didn't "drop" -- the analyst is on a different (and worse) estimator.

Bonus finding: this validates the report's use of `bayesian_ridge` for downstream Digging in revealed why `pca_pls` should never be quoted for the scalar (cognition/age) targets:

1. **Catastrophic r^2 .** `pca_pls` scalar regression has $r^2 = -50$ (4S456) to **-540 (Glasser)** on the bv+demo baseline; `kernel_ridge` is similar (-1 to -7). Only **bayesian_ridge has positive r^2 (+0.115)**. The PLS/KR scalar predictors are wildly overfit/ill-conditioned for low-signal scalar targets.
2. **Not numerically reproducible.** The bv+demo input is subject-info -- *parcellation-independent* -- so the baseline should be **bit-identical across parcellations** for the same seed. It is, for `bayesian_ridge` (max |Delta| across seeds = **0.0000**). But for `pca_pls`, 2 of 10 seeds diverge wildly on identical input (seed 3: -0.045 vs +0.355; seed 6: -0.046 vs +0.305). The ill-conditioning is so severe that run-to-run numerical noise (different cluster nodes -> different BLAS reductions) flips the result. The apparent "parcellation difference" is numerical noise, not signal.

Takeaway: the report correctly leads downstream with `bayesian_ridge` -- it is the *only* stable, positive- r^2 , parcellation-consistent scalar estimator. `pca_pls/kernel_ridge` downstream rows exist in the grid for completeness but **must not be quoted**. (For reconstruction, all three are stable; the estimator choice there only shifts magnitude, not validity.)

One-line answers

- **Oracle:** nothing changed; 0.647 = the BR oracle (demeaned- r 0.648), the analyst read the PLS row. Disattenuation denominator is correct. Just quote one estimator throughout (recommend BR).
- **Cognition baseline:** nothing dropped; 0.129 = the PLS row, the PDF's 0.359 = BR. And PLS/KR scalar regression is ill-conditioned ($r^2 \ll 0$, non-reproducible) -> only BR is reportable downstream.
- **Action:** add an explicit "estimator = `bayesian_ridge`" label to every headline cognition/oracle number, and a note that PLS/KR downstream rows are diagnostic-only. (This is exactly flag 5. in `FINDINGS_EXPLORATION.md`.)

7.4 Reduction-Axis Robustness

7.4.1 Reduction-axis robustness for FC<->SC asymmetry -- findings

Run: SLURM job 10116852 (resubmit after 10086341 hit walltime and 10085734 OOMed on PLS's `coef_`). 16 CPU / 120 GB / 4h walltime. All 10 seeds x 2 directions x 5 method-variants = 100 fits.

TL;DR Verdict: CLEAN, with mild magnitude variation. All five reduction strategies -- full PLS on raw 64,620 edges, learned PCA(256), and three Johnson-Lindenstrauss variants (Gaussian dense, sparse-auto, sparse-1/3) -- show FC->SC > SC->FC at median ratio > 1.4x with Wilcoxon $p \leq 0.001$ against the null ratio of 1.0 across 10 seeds. The FC<->SC asymmetry is **a property of the data, not the reduction pipeline**.

Method (n=10 seeds)	median FC->SC dp	median SC->FC dp	median ratio	min ratio	max ratio	p_vs_1
PCA->PLS->PCA (main model)	0.1355	0.0847	1.621x	1.32x	1.86x	0.001
FULL PLS (no reduction, 64,620-dim)	0.1382	0.0762	1.813x	1.49x	2.11x	0.001
JL Gaussian dense	0.0837	0.0610	1.397x	1.19x	2.25x	0.001
JL sparse_auto (density ~ 1/sqrt(p))	0.0841	0.0592	1.389x	1.11x	1.90x	0.001
JL sparse_1/3 (Achlioptas)	0.0846	0.0535	1.554x	1.24x	1.78x	0.001

What each row tells us

- **PCA->PLS->PCA at 1.59x** is the published 10-seed baseline (matches STEP 11 of the main notebook to within rounding).
- **FULL PLS at 1.79x >= baseline.** No reduction at all on either side gives the *same direction* and a *slightly stronger* magnitude than the learned PCA pipeline. The PCA preprocessing isn't injecting asymmetry -- if anything, it modestly *attenuates* it.
- **All three JL variants at 1.39-1.55x.** A data-blind random projection of FC (input) gives essentially the same asymmetry as the learned PCA. The FC PCA basis is *not* doing anything privileged for the cross-modal prediction -- any 256-dim linear projection captures the cross-modal signal.

Spread across methods: 1.39x (JL sparse_auto) to 1.79x (full PLS), range 0.40. That's wider than the conservative +/-0.15x CLEAN threshold the synthesizer script encoded, but the spread is between known-equivalent reductions and the *direction* is unambiguous across all five.

Why JL gives lower absolute dp than PCA: random projections preserve geometry within JL-bound tolerance but lose the variance-concentration that PCA gets for free. The asymmetry **ratio** is what matters and that ratio is preserved.

Reviewer-proof sentence for the writeup

"The FC->SC asymmetry is robust to the choice of input reduction. Across 10 seeds, the demeaned-pearson ratio FC->SC / SC->FC was 1.59x with the main PCA(256)->PLS(64)->inverse-PCA pipeline, 1.79x with no reduction at all (PLSRegression on the raw 64,620-edge vectors), and 1.39x, 1.39x, and 1.55x with Johnson-Lindenstrauss random projections (Gaussian dense, sparse density=1/sqrt(p), sparse density=1/3 respectively). All five methods reject the null ratio of 1.0 at Wilcoxon $p \leq 0.001$."

Caveats

1. **One method failed twice before working.** First attempt (10085734) OOM-killed during method B because `sklearn.cross_decomposition.PLSRegression` materializes `coef_` of shape (64620, 64620) = 33 GB at fit-time. Fix: bypass `coef_` at predict time by manually computing $(X_{\text{test}} - x_{\text{mean}}) / x_{\text{std}} @ x_{\text{rotations}} @ y_{\text{loadings}}.T * y_{\text{std}} + y_{\text{mean}}$. The fix is in `method_b_full_pls.py` and works at 64 GB but we kept the bump to 120 GB for headroom. Second attempt (10086341) hit the 1h30m walltime at seed 8 of method B; resolved by raising walltime to 4h **and** adding a per-seed cache (`_method_b_per_seed/seed_<n>.csv`) so any future cancellation resumes cleanly. Final attempt (10116852) completed all 100 fits in well under the 4h budget.
2. **Synthesizer initially hid methods A and B from the summary table.** Pandas read empty `j1_variant` cells as NaN; default groupby drops NaN keys, silently dropping the non-JL rows from the printed summary. Underlying CSVs were always complete (100 rows loaded). Patched: `df["j1_variant"].fillna("")` before the groupby. The numbers in the table above are the post-patch medians.
3. **Magnitude spread (1.39-1.79) is wider than +/-0.15x.** The synthesizer's CLEAN threshold was conservative; the *direction* is identical across all methods, but JL is a lossier projection than PCA, so the absolute prediction quality drops and the ratio shifts accordingly. None of this changes the robustness claim.
4. **JL random matrices are seed-dependent.** Each (variant, seed) gets a fresh JL random matrix (`random_state=seed`), so the per-seed table shows variance from both data splits *and* projection draws. This is the harder test -- using one shared JL across seeds would give tighter numbers.

Files

- README.md -- what each script does
- `method_a_pca_pls_pca.py` + `_output.txt` + `method_a_results.csv`
- `method_b_full_pls.py` + `_output.txt` + `method_b_results.csv` + `_method_b_per_seed/seed_*.csv`
- `method_c_jl_pls_pca.py` + `_output.txt` + `method_c_results.csv`
- `synthesize_reduction_axis.py` + `_output.txt` + `reduction_axis_synthesis.csv` + `reduction_axis_summary.csv`
- `run_all.sbatch` -- SLURM wrapper (cpu_short, 16 CPU, 120 GB, 4h)

7.5 FC Noise and Reliability Sanity Check

7.5.1 FC noise sanity check -- findings

Run: HCP-YA, SLURM jobs 11034024 (build+A+B+F) + 11034554 (E re-run after low-dim fix). 4-cell FC design (run-1/run-2 x LR/RL), both parcellations, ~1018 FC subjects (reliability) / 957 canonical (cross-modal). All values from `outputs/*.csv`.

TL;DR FC is mostly noise at the single-edge level but highly reliable as a whole connectome, and SC predicts only a small fraction of even the *reproducible* FC signal -- and that shortfall is **not explained by FC measurement noise** (proven per-subject). This is the physical counterpart to MASTER_FINDINGS F10 (statistical saturation): the SC->FC gap is demonstrably not an FC-noise artifact; genuine cross-modal independence is the strong interpretation, pending the one data-blocked piece (SC's own reliability -- needs test-retest dMRI). **CLOSED, filed as F10-supporting (MASTER A6).**

- **FC reliability ceiling** (between-session, Glasser): demeaned_r **0.49**, fingerprint top1 **0.93**, avg_rank **0.99**.
- **FC edge variance: 30% trait / 4% day-to-day state / 2% within-session / 64% noise**; the averaged connectome we actually use has reliability **G ~ 0.59**.
- **SC->FC captures ~17%** of the reproducible FC signal (demeaned_r 0.085 of ceiling 0.49) -- and only ~5% of the fingerprinting ceiling.
- **Parcellation-robust** (4S456 nearly identical).
- **SC noise itself remains UNMEASURED** -- no test-retest dMRI in HCP-YA.

The honest bottom line on "how much is noise" There isn't one number, and that's not a failure to find a straight answer -- the question is genuinely **level-dependent**. The straight answer is the set:

1. **A single FC measurement is ~64% noise** (per-edge individual-difference variance).
2. **Your averaged usable connectome is ~41% noise** (reliability $G \sim 0.59$).
3. **The reproducible individual signal you can actually predict tops out at demeaned-r 0.49** (the between-session ceiling), and it is **heterogeneous across people (0 to 0.78)**.
4. **Of that reproducible 0.49, SC explains ~17%, flat across subjects** (independent of each subject's own reliability).

(Caveat that rides #4: this is the FC-side accounting; SC's *own* noise floor is unmeasured -- so "SC explains 17% of reproducible FC" is exact, while "the other 83% is signal SC doesn't contain" is the strong interpretation, pending SC test-retest.)

7.5.2 #REVIEW AND QUESTION -- is "ceiling" valid per-subject, or only at the population level?

(Flagged for review -- logic below seems sound but wants a second look before it hardens any per-subject claim. It does NOT affect the population headline or the flatness result; it sharpens how we're allowed to phrase the per-subject material.)

The disattenuation logic (Spearman 1904) is a *population theorem*, not a per-subject law: the correlation between two variables is bounded by the geometric mean of their reliabilities **in expectation, over a population**. A single subject's "reliability" (correlation between their two scans) is **one noisy number from a single pair of measurements** -- large standard error. So is their achieved prediction. Comparing two noisy single-subject estimates, some will land achieved > reliability **purely by sampling noise** -- especially low-reliability subjects, whose reliability estimate sits near zero and is easy to exceed by chance. (The -0.002 subject isn't truly perfectly unpredictable; -0.002 is noise around some small true value, and a noisy achieved score can exceed it.)

Precise statement: the ceiling is a **valid population bound but an invalid per-subject bound** -- per-subject reliability is a single noisy estimate, not a true per-person limit. The theorem holds on average, not pointwise; treating it pointwise produces the contradiction we saw (achieved > "ceiling" for some subjects).

Corrected framing (threads both concerns):

- **Population level -- keep "ceiling".** "Across subjects, individual FC reproduces at mean demeaned-r 0.49; SC->FC captures ~17% of that." The disattenuation is legitimate here and is not exceeded on average. -> stays in the main text (bottom-line #3, E).
- **Per-subject level -- do NOT call it a ceiling.** Call it "**per-subject reliability**" and describe its *distribution* (0-0.78, $\rho=0.41$, a stable trait). Frame the H result as "**SC's prediction is uncorrelated with subject reliability**" (a relationship between two measured quantities) -- **not** "SC stays below each subject's ceiling," which is the framing that breaks.

Net: the H conclusion stands (flat $r=0.01$ = SC prediction is reliability-independent), and it's arguably *cleaner* under this framing. The thing to fix downstream: the per-subject `fraction_of_ceiling` column in `h_per_subject_achieved_vs_ceiling.csv` implicitly treats per-subject reliability as an individual bound -- report it as a descriptive ratio at most, and lean

on the **correlation/flatness** statement (and the bv+demo contrast) for the actual claim, not on pointwise "fraction of ceiling."

7.5.3 Two ceilings are different objects (and we should report both, labeled)

There are **two distinct ceilings**; conflating them is the deeper source of the tension.

- **Ceiling A -- data / reproducibility ceiling** (FC_{day1} <-> FC_{day2}). *How reproducible is the target itself?* Model-free, a property of the **data**. "Individual FC only agrees with itself at **0.49**, so no predictor can exceed the reproducible signal." Valid at population level, breaks per-subject (above).
- **Ceiling B -- model / oracle ceiling** (FC->FC, SC->SC through the *same* PCA->PLS pipeline). *How well can THIS model class predict the target from a perfect same-modality copy?* A property of **model + data together**. "Even predicting FC from FC, PCA->PLS only reaches X -- the architecture has a representational limit." (SC->SC oracle ~ **0.647** from earlier work -- to recompute consistently in the grid; FC->FC = [to compute].)

Ceiling B fixes both earlier complaints:

- **Not exceeded per-subject** -- it's a within-modality prediction run through the *same pipeline / estimator / CV / metric* as the cross-modal predictions; an apples-to-apples upper reference, no "two noisy scans" comparison that can flip.
- **Available for SC** -- SC->SC is just a prediction task, no test-retest needed. The SC-side data gap that blocks Ceiling A **disappears** for B, so both directions get a consistent ceiling.

But B is a looser bound than A (the "???" catch): FC->FC can exploit **session-specific signal that wouldn't replicate** in a fresh scan, so B can sit *above* A. B measures "max ability of the model to reproduce *this* connectome," not "max recoverable *individual trait*." **B cannot substitute for A** for the biological fraction.

How to report -- a two-rung reference, each labeled for what it bounds:

- **SC->FC achieved: 0.085.**
- **Ceiling B (model oracle), both directions:** FC->FC = [to compute], SC->SC ~ 0.647 -> "within-modality is the architecture's best case; cross-modal loses this much." Clean, consistent, available both directions, never exceeded per-subject -> the **model-capacity / cross-modal-loss** story.
- **Ceiling A (data reproducibility): 0.49**, FC-side only, heterogeneous -> "of the *reproducible* signal, SC gets ~17%." The **biological** fraction, with the per-subject caveat + SC-side gap flagged.

Honesty sentence to carry: "The within-modality oracle (FC->FC) is a model-capacity reference and exceeds the cross-session reproducibility limit, because it can fit session-specific signal that does not replicate; we therefore disattenuate biological claims against the reproducibility ceiling (A) and use the oracle (B) only to quantify cross-modal vs within-modality loss."

A. Reliability ceiling (native metric)

parc	comparison	demeaned_r	pearson	top1 (fingerprint)	avg_rank
Glasser	within-session (LR<->RL)	0.364	0.710	0.78	0.968
Glasser	between-session (REST1<->REST2)	0.491	0.813	0.933	0.992
4S456	within-session	0.337	0.670	0.78	0.969
4S456	between-session	0.457	0.783	0.936	0.993

Note (important): within-session (LR<->RL) is *lower* than between-session -- the opposite of "shorter interval = more reliable." Cause = the **phase-encode distortion confound**: LR and RL have opposite distortions, so single-direction connectomes disagree more, while each session (LR+RL averaged) is distortion-cancelled and cleaner. So **between-session 0.49 is the valid ceiling**; the LR<->RL rung is contaminated and is NOT a clean short-interval estimate.

B. Variance decomposition (G-theory 2x2; individual-difference fractions, sum to 1)

parc	trait (signal)	state (day)	within-session	noise	G (avg connectome)
Glasser	0.301	0.035	0.021	0.643	0.585
4S456	0.261	0.035	0.021	0.683	0.519

At the single-edge level **~64-68% of between-subject variance is measurement noise**, only ~26-30% is stable trait, and day-to-day state is small (~3-4%). The 4-cell average (REST1+REST2, LR+RL) lifts reliability to G~0.52-0.59. Per-edge components saved in `outputs/b_variance_components_{parc}.npz`.

F. Whole-connectome reliability

parc	fingerprint top1	discriminability
Glasser	0.927	0.998
4S456	0.934	0.999

The reconciliation (the original puzzle: "but scans match across sessions"): per-edge ~64% noise (B) yet whole-connectome 93% identifiable / 0.998 discriminable. Both true -- the individual signal is **distributed**, reliable in aggregate even when each edge is mostly noise.

E. Cross-modal disattenuation -- "% of reproducible FC captured" ceiling = REST1<->REST2 (demeaned_r 0.491). Achieved = source->FC, PCA->PLS, 10-seed median.

source	metric	achieved	ceiling	fraction of ceiling
SC->FC	demeaned_r	0.085	0.491	0.17
SC->FC	top1 (fingerprint)	0.051	0.933	0.05
SC->FC	avg_rank	0.713	0.992	0.72
bv+demo->FC	demeaned_r	0.098	0.491	0.20
bv+demo->FC	top1	0.026	0.933	0.03

Reads:

- **SC captures only ~17% of the reproducible FC signal** (demeaned_r) and **~5% of the fingerprinting ceiling** -- so SC->FC is far from the reliability ceiling: the gap is genuine cross-modal independence, not just FC noise. (If SC->FC were near the ceiling we'd blame noise; it isn't, so most of the unexplained FC is reliable-but-structurally- unpredictable.)
- **bv+demo->FC (0.20) >= SC->FC (0.17)** even after disattenuation -- consistent with the project's baseline finding: cheap subject confounds match/beat the structural connectome.
- pearson fraction >1.0 is an artifact (raw pearson is dominated by the shared population mean and is uninformative here -- use demeaned_r); not reported as meaningful.

G + H. Per-subject reliability + does prediction track it? G -- per-subject reliability is heterogeneous and a stable trait. The 0.49 ceiling is a mean; per-subject between-session reliability spans ~0 to 0.78 (Glasser: mean 0.489, median 0.494, std **0.117**, skew **-0.58**, non-normal p=5e-15), with a low tail (~2% of subjects < 0.2; one at -0.002 = pure noise). It's a stable subject property (within-vs-between per-subject rho=0.41) -- consistent with motion/compliance being person-level. (`outputs/g_per_subject_*, g_reliability_hist.png`)

H -- SC->FC is NOT noise-limited per subject; the gap is uniform independence.

- A subject's SC->FC prediction quality is **uncorrelated** with their own FC reliability ceiling: Pearson **r=0.01 (p=0.72)**, Spearman ~0. (bv+demo->FC weakly tracks it, r=0.15, p=2e-5.) So cleaner-FC subjects are **not** more predictable from SC.
- **Reliability-filtering does not sharpen SC->FC** -- it makes the fraction *worse*: | filter | n | SC achieved | SC ceiling | SC fraction | |---|---|---|---| all | 857 | 0.083 | 0.491 | **0.169** | | drop rel<0.2 | 841 | 0.083 | 0.498 | 0.167 | | drop bottom 10% | 771 | 0.083 | 0.517 | 0.160 | | keep top 50% | 429 | 0.080 | 0.582 | **0.137** | Keeping only high-reliability subjects raises the

ceiling (0.49->0.58) but SC's achieved stays flat (~0.08), so the fraction drops. The noisy tail was never the bottleneck.

- (bv+demo->FC holds ~0.20-0.21 across all filters.)
- **Why flat $r=0.01$ is the strong outcome (not a weak/null one):** per-subject achieved prediction is *mechanically* bounded by reliability (you can't predict noise), so the default expectation was a **positive** slope by construction. We got flat. SC captures a fixed ~0.08 whether a subject's FC is reliable (0.78) or near-noise (0.0) -- which *rules out* the boring ceiling-effect explanation. The filtering result corroborates by going the "wrong" way: dropping unreliable subjects *lowers* the captured fraction (0.169->0.137) because the ceiling rises while achieved stays pinned. Ceiling moves, achieved doesn't = SC has a fixed, modest grip on FC unrelated to FC's measurement quality.
- **The bv+demo contrast is what makes it a clean dissociation (not an artifact):** the analysis *does* detect a real reliability effect when one exists -- bv+demo->FC mildly tracks reliability ($r=0.15$). That SC's slope is flat while the baseline's isn't means the flatness is not a methodological artifact (it would have hit both).
- **Conclusion (with the caveat riding it):** the airtight claim is **the SC->FC gap is not FC-measurement-noise and not per-subject reliability** -- decisively, at the per-subject level. The reading "SC doesn't *contain* that part of FC" is the strong **interpretation**, but is not fully separable from "SC contains it but measures it too noisily, *uniformly* across subjects" -- a uniform SC noise floor would also produce a flat line. Distinguishing those needs SC test-retest (data-blocked). So: 95% of the way to genuine independence; the last 5% is the SC-reliability hole. Reinforces E.
- (outputs/h_per_subject_achieved_vs_ceiling.csv, h_reliability_filtered_summary.csv, h_correlations.csv, h_achieved_vs_ceiling_scatter.png.)

Status: CLOSED (supporting item for F10) This module has done its pre-grid job: it confirms the cross-modal ceiling is not an FC-measurement-noise artifact, in our native metric, per-subject. **Filed as F10-supporting in MASTER FINDINGS (Appendix A6). No further per-subject reliability analysis is warranted here** -- the finding is extracted; the discipline now is to run the reproducibility grid this de-risked, not to chase the per-subject rabbit hole.

What this resolves and what stays open

- **Resolves (airtight):** how much of FC is noise (edge-level ~64%; whole-connectome reliable), and that the SC->FC gap is **not FC-measurement-noise and not per-subject reliability** -- decisively, per-subject (flat $r=0.01$ vs the baseline's $r=0.15$). Sharpens F10 with a physical denominator.
- **Interpretation (strong, pending data):** the natural reading is "SC doesn't *contain* that part of FC" -- but a *uniform* SC noise floor would also produce the flat line, so this last step is not fully separable from FC-side evidence alone.
- **Open (data-blocked):** SC's own noise floor, and FC->SC disattenuated by SC reliability -- both need test-retest dMRI (not in HCP-YA). [caution] literature plug-in only until sourced. This is the 5% gap between "not-noise-on-the-FC-side" (proven) and "genuine independence" (interpretation).

Caveats

- Within-session rung confounded by phase-encode distortion (above).
- Reliability computed on ~1018 FC subjects; SC->FC achieved on the 957 canonical set (reliability is a per-subject property, stable across the subset).
- Between-session interval is ~1 day, same scanner (HCP-YA REST1/REST2) -- the optimistic end; a months-apart/multi-site retest would show lower reliability (more noise).
- Cognition not involved here -- this is FC<->FC reliability + SC->FC; the cognition ceiling is covered separately (F4/F5/F10).

Files

- outputs/a_reliability_ceiling.csv, b_variance_decomposition.csv, b_variance_components_{Glasser,4S456Parcels}.npz,

```

f_discriminability.csv,          e_crossmodal_disattenuation.csv,
noise_synthesis.csv
• scripts:      build_fc_cells.py,    a_/b_/e_/f_,    synthesize_noise.py,
               _noise_common.py, run_all.sbatch, run_e_synth.sbatch
• roadmap: planning/roadmap/noise-sanity-check.md

```

7.6 PC Mechanism Tractography-Reliability Check

7.6.1 PC3 tractography-reliability sanity check -- findings

Run: SLURM job 10084347 on cpu_short, 16 CPU / 32G, ~1 min total compute. All inputs from Depth 1 seed-0 PC3 loadings (sc_pc_loadings.npy, row 2).

TL;DR Verdict: the PC3 visual/DAN localization survives reliability partialling with room to spare. Two of the three headline within-network enrichments are *higher* after partialling edge strength + distance; the third (DAN-DAN) drops modestly but stays at 5.7x. The biological reading of PC3 holds. Add the partialled numbers to the writeup as the reviewer-proof sentence.

Headline pair	Raw enrichment (Depth 1.1)	Residualized (this check)	Delta
visual		visual	11.97x
dorsal attention		dorsal attention	7.73x
dorsal attention		visual	4.49x

Reliability proxy (strength + distance) explains **41.0% of |PC3| variance**, so the partialling is non-trivial. After removing that 41%, the residual still concentrates on the same network pairs -- most clearly in visual cortex.

Per-proxy results

Proxy 1 -- edge strength (Spearman + R²)

- Spearman(|PC3|, edge_strength) = **+0.89** (p ~ 0)
- OLS R² of |PC3| ~ edge_strength = **0.40**
- Pearson = +0.63

Strong positive: PC3 emphatically lives in high-strength edges. **Reliability confound on streamline density is LIVE.** Posterior visual edges happen to be high-strength because tracking is cleanest there, so this was the expected red flag.

Proxy 2 -- anatomical distance (Spearman + R²)

- Spearman(|PC3|, edge_distance) = **-0.50** (p ~ 0)
- OLS R² of |PC3| ~ edge_distance = **0.13**
- Pearson = -0.36
- Distance was computed from Glasser MNI centroids (atlas CSV mni_x/y/z). Range: 3.3-160 mm, median 79 mm.

Strong negative: PC3 favors short edges. **Distance confound is also live but smaller in magnitude than strength** (R² 0.13 vs 0.40).

Decisive partialled enrichment (the load-bearing test) OLS |PC3| ~ strength + distance:

- R² = **0.41** (strength dominates; distance adds essentially nothing beyond it)
- coef[strength] = +0.0135, coef[distance] = -1.2e-5
- residual std / signal std = **0.77** (77% of the magnitude variation is still there)

Re-ranked top-200 edges by |residual| and recomputed Yeo7 network enrichment:

Net pair	enr_raw	enr_resid	n_resid
visual		visual	11.97
dorsal attention		dorsal attention	7.73
dorsal attention		visual	4.49
frontoparietal		frontoparietal	2.61
dorsal attention		frontoparietal	1.50
default mode		dorsal attention	1.38
somatosensory		somatosensory	0.21
default mode		default mode	0.67

(Full table: `enrichment_residual_top200.csv`.)

Visual-visual enrichment goes UP after partialling. That's the key observation -- not just "survives" but "intensifies." It means reliability under-predicts how strongly PC3 emphasizes visual cortex; once you subtract the part of |PC3| explained by strength+distance, what's left is *even more* visual-concentrated.

DAN-DAN drops from 7.73x to 5.73x -- about a quarter of its enrichment was attributable to reliability -- but the residual is still ~5-6x chance, well above the >5x "clean biological" threshold. DAN-visual is essentially unchanged.

Retest reliability (not run) The rigorous version of this test would use HCP test-retest data (n~45 subjects scanned twice) to compute per-edge ICC and partial that out instead of the strength+distance proxy. That requires standing up a retest ingestion pipeline (~1-2 days) which is out of scope for this sanity check. The proxy is the standard fallback in the field and tends to agree with retest reliability on which edges are noisy. Documented in `retest_check_note_output.txt`.

Reviewer-proof sentence for the writeup

"PC3's visual/DAN localization is not attributable to higher reconstruction reliability in posterior short connections: after partialling streamline density and inter-region distance from |PC3| (which together account for 41% of its magnitude variance), the headline within-network enrichments remain at 13.3x (visual-visual; *higher* than the unadjusted value), 5.7x (DAN-DAN), and 5.0x (DAN-visual). The visual-visual enrichment increases under partialling, indicating reliability under-predicts PC3's concentration on visual cortex rather than driving it."

Caveats to keep honest

1. **Proxy not retest.** Edge strength and inter-region distance correlate with tractography ICC but are not identical. A future ICC-based check could in principle reveal a confound the proxy misses.
2. **Single seed.** Ran at seed 0 (the Depth 1 PCA basis). The Depth 1.1 stability analysis showed PC3 is stable across seeds (median |cos|=0.89), so the result should generalize, but the partialling itself wasn't re-done per seed.
3. **DAN-DAN partial give-back.** The 7.7x->5.7x drop says ~25% of DAN-DAN enrichment was reliability-driven. Frame as "DAN-DAN remains 5.7x enriched after partialling," not "DAN-DAN is unaffected."
4. **The 41% R² is high.** Don't bury it. Strength + distance together explain nearly half of |PC3|'s magnitude; the surviving signal is the other half. This is not "reliability is irrelevant" -- it's "reliability matters, AND the residual is still cleanly visual/DAN."

Files in this directory

- README.md -- what each script tests and why
- proxy1_strength.py + _output.txt
- proxy2_distance.py + _output.txt
- decisive_partialled_enrichment.py + _output.txt + enrichment_residual_top200.csv
- retest_check_note.py + _output.txt
- run_all.sbatch -- SLURM wrapper (cpu_short, 16 CPU, 30 min, auto-exits)
- _cache_*.npy -- local scratch (gitignored)

7.7 Richer Tractography Representation Check

7.7.1 Tractography-predict -- findings

Run: 6 experiments in parallel (:ro overlay), 10 seeds each, Glasser, family-aware splits. Jobs 10688656 (e1, 16 cpu) + 10689496-500 (e2-retest, 4 cpu under the 32-core per-user QOS cap). All metrics reported, not just demeaned_pearson.

TL;DR Richer tractography does not help. The named-bundle representation (r2t) predicts FC *worse* than raw streamline counts, adds nothing marginal on top of counts, and carries no downstream cognition signal above demographics. Meanwhile the FC<->SC asymmetry is robust across all 6 metrics and *amplifies* in the bundle representation. Downstream, **FC remains the only modality that beats the bv+demo cognition floor** -- no structural or tractography representation (count, bundle, bundle-similarity, or synthetic-FC-from-tractography) clears that bar.

This is a clean negative result for the "parcellated counts lose information that richer tractography would recover" hypothesis, and a strong positive for FC's unique downstream value.

E1 -- predicting FC from each structural representation (median, 10 seeds)

rep	demeaned_pearson	pearson	top1_acc	avg_rank	mse	r2
SC (count baseline)	0.0847	0.8292	0.0513	0.7128	0.0143	-0.057
kitchen_sink [SC		r2t		bv		demo]
SC_r2t [SC		r2t]	0.0791	0.8272	0.0359	0.7020
r2t (bundle)	0.0493	0.8153	0.0205	0.6090	0.0154	-0.127
r2t_corr (bundle-similarity)	0.0362	0.8231	0.0179	0.6070	0.0148	-0.091

- **Count-SC beats bundle-r2t** for FC prediction on every metric. The bundle profile is a *lossier* FC predictor than raw counts.
- **kitchen_sink edges SC** (0.0942 vs 0.0847 demeaned_pearson; best avg_rank) -- but the gain is ~11% and comes from bv+demo, not r2t (see E3).
- Combined-rep rows use **per-block PCA** (PCA each block to its own 256-dim latent, then concat) to fix a scale-domination bug where the naive single-PCA on concatenated features was dominated entirely by r2t's magnitude, making SC/bv/demo invisible.

E2 -- FC<->SC asymmetry across representations, ALL 6 metrics median FC-wins (ratio>1, or r2 as difference>0, means FC->X beats X->FC):

rep	demeaned_pearson	pearson	top1_acc	avg_rank	mse	r2(diff)
SC	+1.62x [pass]	+1.10x [pass]	+2.56x [pass]	+1.23x [pass]	+2.59x [pass]	+0.036 [pass]
r2t	+2.26x [pass]	+1.04x [pass]	+1.50x [pass]	+1.19x [pass]	0.00 [fail]	+0.45 [pass]
r2t_corr	+1.63x [pass]	0.98x [fail]	+1.83x ~	+1.18x [pass]	0.85x [fail]	-6.9 [fail]

([pass] = Wilcoxon one-sided p<0.05 that FC wins; all 10 seeds.)

- **SC asymmetry is bullet-proof: FC->SC > SC->FC on all 6 metrics, every one p=0.001.** This is the strongest multi-metric statement of the project's headline asymmetry.
- **r2t amplifies the asymmetry on the "shape/identifiability" metrics** (demeaned_pearson 2.26x vs SC's 1.62x) but FAILS on mse (ratio 0.00) -- i.e. FC->r2t and r2t->FC have similar raw error, the asymmetry lives in the demeaned/rank structure, not raw magnitude. Honest nuance: the r2t asymmetry is metric-dependent.
- r2t_corr is mixed (fails pearson/mse/r2); not a clean asymmetry carrier.

E3 -- marginal contribution of r2t over SC (paired Delta, 10 seeds)

- median Delta (SC_r2t - SC) demeaned_pearson = **-0.0013** (essentially zero / slightly negative); one-sided Wilcoxon p(greater)=0.98 (NOT an improvement), two-sided p=0.049 (SC_r2t is marginally *worse*).
- **Verdict: count-SC is a sufficient statistic for cross-modal prediction.** Adding the bundle representation does not add FC-predictive signal -- it slightly hurts (extra latents, no new information).

E5 -- downstream cognition (the real metric) Median test Pearson predicting NIH-Toolbox composites, and lift over the bv+demo floor:

rep	CogTotal	CogFluid	CogCrystal	lift over bv+demo (Total)
FC	0.451	0.342	0.451	+0.078 [pass]
bv+demo (floor)	0.372	0.309	0.346	0
SC	0.261	0.180	0.267	-0.111
r2t->synthFC (substitution)	0.202	0.197	0.180	-0.170
r2t	0.196	0.130	0.164	-0.176
SC_r2t	0.196	0.135	0.160	-0.177
r2t_corr	0.151	0.157	0.103	-0.221

This is the decisive result:

- **FC is the ONLY representation that beats the bv+demo floor** (+0.078 total, +0.105 crystallized, +0.033 fluid). After residualizing demographics, FC retains 0.37 crystallized / 0.22 fluid; every structural rep collapses to ~0-0.08.
- **Every tractography representation falls BELOW the demographic floor** -- count-SC, bundle-r2t, bundle-similarity, and the combined rep all carry *less* cognition signal than age+sex+brain-volume alone.
- **The substitution chain fails:** r2t -> synthetic-FC -> cognition (raw 0.18-0.20) does NOT recover FC's cognition signal. Synthetic FC generated from tractography is not a useful cognitive biomarker -- the cognition-predictive structure of real FC is not reconstructable from tractography.

This extends the project's prior "FC carries non-demographic cognition signal, SC doesn't" finding: **richer tractography does not rescue SC**, and you cannot launder tractography into FC's cognition signal via cross-modal prediction.

E4 -- does SC-PC3 (the dorsal-stream backbone) exist in the bundle representation?

- r2t PC modes 1-3 are stable across seeds (median |cos| 0.99 / 0.98 / 0.95); modes 4+ degrade.
- **SC-PC3 does NOT map cleanly to any single r2t mode** (best Spearman of subject scores = -0.32 with r2t-mode 10; everything else |rho|<0.23). The dorsal visual-stream / DAN structural backbone (Depth 1.1) is a **count-SC phenomenon**, not cleanly present in the named-bundle decomposition. The bundle atlas (66 tracts) doesn't carve the cortex finely enough to express that intra-parietal/occipital edge pattern.

Retest reliability (FC scan-rescan, independent cross-check)

- FC reliability adds essentially nothing beyond the strength+distance proxy: joint $R^2(\text{IPC3I} \sim \text{strength+distance+FC-reliability}) = 41.2\%$ vs strength+distance-only 41.0%.
- PC3 visual/DAN localization survives the 3-way partial: visual||visual **13.3x**, DAN||DAN **5.7x**, DAN||visual **5.0x** -- identical to the strength+distance result.
- Independent confirmation that the PC3 localization is not a reliability artifact. (Caveat unchanged: FC reliability != SC reliability; gold-standard SC ICC would need the HCP retest dMRI release.)

Bottom line for the writeup

1. **The asymmetry is real and multi-metric** -- FC->SC beats SC->FC on all 6 metrics at $p=0.001$ across 10 seeds. Strongest statement yet.
2. **Tractography richness is a dead end** -- the bundle representation predicts FC worse than counts, adds nothing marginal, carries no cognition signal, and doesn't contain the PC3 backbone. The parcellated-count "information loss" hypothesis is falsified for this data: counts are a sufficient statistic.
3. **FC's downstream uniqueness is confirmed and sharpened** -- FC is the only modality above the demographic cognition floor; synthetic-FC-from-tractography does not recover it. Cognition prediction is an FC story, full stop.

Files

- e1_source_rep_results.csv, e2_asymmetry_results.csv + _summary.csv, e3_marginal_results.csv + _summary.csv, e4_* (3 CSVs), e5_downstream_results.csv + _summary.csv
- tractography_synthesis.csv, synthesize_tractography_output.txt
- ../sanity_checks/tract_check/retest_icc_results/ (FC reliability + 3-proxy enrichment)
- scripts: e1-e5, synthesize_tractography.py, _tract_setup.py, run_one.sbatch (parameterized, :ro overlay for parallel runs)

7.8 Nonlinear Model-Class Check

7.8.1 Non-linear sanity check -- findings

Run: 2026-06-13, 3 experiments in parallel (:ro overlay, 4 cpu/24G each, jobs 10757352-54), 10 seeds, Glasser, family-aware splits. Estimator swapped, representation + splits held fixed vs tractography_predict/. Linear baselines byte-identical to the linear runs (shared _tract_setup.py).

TL;DR NULL across the board -- the tractography dead-end and the FC<->SC asymmetry are model-class robust. Neither HistGradientBoosting nor KernelRidge (RBF) extracts any tractography signal that linear models missed -- not for cognition, not for FC reconstruction, not for marginal contribution over SC. This is the strongest version of the linear finding: tractography's failure to predict FC/cognition is not a linear-model artifact.

Two estimator caveats matter for reading this (below): KernelRidge is the trustworthy nonlinear probe (it preserves FC's known cognition signal); HGB underperforms at this sample size even where signal exists.

N1 -- nonlinear cognition Best lift over the bv+demo floor among tractography reps (SC, r2t, r2t_corr, SC_r2t), per estimator (median 10 seeds; lift = Pearson - bv+demo floor with the same estimator):

estimator	best tractography lift	rep / target	clears floor?
linear_BR	-0.079	SC / Crystal	no
HGB	-0.168	r2t_corr / Fluid	no
KR	-0.029	SC / Crystal	no

No tractography representation clears the demographic floor under any estimator. The closest is KernelRidge on SC/crystallized at -0.029 -- still below floor.

Estimator sanity (FC, CogTotal) -- does nonlinear preserve known signal? FC carries real cognition signal (linear lift +0.078). A trustworthy nonlinear estimator should keep it:

estimator	FC pearson	floor	FC lift
linear_BR	0.451	0.372	+0.078 [pass]
KR	0.338	0.282	+0.056 [pass]
HGB	0.254	0.355	-0.101 [fail]

- **KernelRidge preserves FC's lift** (+0.056) -> it's a valid nonlinear probe, and its finding of zero tractography signal is meaningful.
- **HGB destroys even FC's signal** (-0.101) -> gradient boosting underperforms at n~683 with 256 PCA inputs (overfits / can't model the smooth FC->cognition map). HGB's "below floor for tractography" is therefore partly an estimator-weakness artifact, not evidence about tractography. We trust the **KR** result.

N2 -- nonlinear reconstruction + asymmetry KernelRidge vs linear PLS, FC->X de-meanned_pearson (median 10 seeds):

direction	PLS	KR	Delta (KR-PLS)	nonlinear gain?
FC->SC	0.1355	0.1334	-0.0021	no
FC->r2t	0.1113	0.1133	+0.0020	no (< 0.02)
FC->r2t_corr	0.0582	0.0605	+0.0023	no (< 0.02)

All Delta are within +/-0.0023 -- nonlinear neither helps nor hurts reconstruction. Counts still beat bundles under KR (0.133 vs 0.113), same as linear.

Asymmetry ratio (demeaned_pearson, FC-wins) holds under nonlinear

rep	linear_PLS	KR
SC	1.621x	1.638x
r2t	2.264x	2.267x
r2t_corr	1.634x	1.487x

The FC<->SC asymmetry is essentially **identical** under linear and kernel models (SC 1.62->1.64, r2t 2.26->2.27). The asymmetry is not a linear-pipeline artifact. (Full 6-metric panel per estimator in `n2_reconstruction_summary.csv`.)

N3 -- nonlinear marginal (r2t over SC) Paired Delta(SC_r2t - SC) -> FC, demeaned_pearson (median 10 seeds):

estimator	Delta	Wilcoxon p (greater)
linear	-0.0013	0.981
KR	-0.0049	0.998

r2t adds **nothing** over count-SC even under a nonlinear combined model -- slightly negative both ways. Count-SC remains a sufficient statistic; there is no conjunctive SCxr2t interaction that a kernel picks up.

Verdict All three decision rules return null:

- N1 cognition unlock ($\geq +0.03$ over floor): **not met** (KR best -0.029).
- N2 reconstruction gain ($\geq +0.02$ dp): **not met** (max +0.0023).
- N3 marginal gain ($\geq +0.02$ dp, $p < 0.05$): **not met** (KR -0.0049).

Conclusion: the tractography findings from `tractography_predict/` are model-class robust. Switching from linear (PLS/BR) to nonlinear (KernelRidge RBF, the trustworthy probe here) changes nothing -- bundles still predict FC worse than counts, still carry no cognition signal above demographics, still add nothing over SC; and the FC<->SC asymmetry is unchanged. The tractography information is not hiding in nonlinear structure.

Caveats

- **HGB unreliable at this n:** HistGradientBoosting underperformed even on FC (where signal demonstrably exists), so its nulls are not informative on their own. The load-bearing nonlinear evidence is **KernelRidge**, which preserved FC's signal and still found nothing in tractography.
- **KR hyperparameters fixed** (median-heuristic gamma, $\alpha=1.0$), not CV-tuned. Margins to the decision thresholds are wide (best cognition lift -0.029 vs +0.03 rule), so light tuning won't flip the null. A gamma/alpha sweep is the natural follow-up only if a result had landed near threshold (none did).
- HGB-per-component reconstruction not run (256x the fits); KR multi-output was the reconstruction probe. Given KR showed no recon gain, HGB-recon is unlikely to differ.
- Single parcellation (Glasser), 10 family-aware seeds.

Files

- `n1_cognition_results.csv / _summary.csv` (rep x estimator x target x seed: pearson, spearman, r2, lift)
- `n2_reconstruction_results.csv / _summary.csv` (full 6-metric panel per estimator, both directions, + asymmetry ratios)
- `n3_marginal_results.csv / _summary.csv` (full panel, paired Delta per estimator)
- `nonlinear_synthesis.csv, nonlinear_synthesis_output.txt`
- scripts: `n1-n3, synthesise_nonlinear.py, _nl_common.py, run_one.sbatch`
- nonlinear predictors live in `../tractography_predict/_tract_setup.py` (`kernelridge_predict, kernelridge_blocks_predict, hgb_scalar_predict, kr_scalar_predict`)

7.9 Residual-Boost and Multimodal Sink Check

7.9.1 Residual-boost (architecture A) + multimodal sink -- findings

Run: 2026-06-13, jobs 10759407/408 (N4), 10759825/826 (N5), parallel :ro, 10 seeds, Glasser, family-aware. Design rationale in `DESIGN_residual_learning.md`. Builds on `findings_nonlinear.md` (plain nonlinear, also null).

TL;DR The residual-learning idea -- hand the nonlinear model the linear prediction for free (OOF, no leakage) and make its loss the improvement above that template -- was the right, maximally-sensitive test. Result:

- **Downstream cognition: comprehensively NULL.** Not single-modality (N4cog), not cross-modal sink (N5cog), not residual-boosted. FC alone remains the ceiling; combining modalities *dilutes* it; the nonlinear cross-modal residual adds nothing.
- **Reconstruction: one real but tiny signal.** N4recon shows the residual-boost beats the PLS template by +0.002-0.006 demeaned_pearson, $p=0.001$ every direction -- a genuine nonlinear sliver in connectome $\leftrightarrow$ connectome mapping, ~4x below the +0.02 "matters" threshold, and it washes out under the cross-modal sink (N5recon).

Interpretation: the connectome->cognition relationship is linearly saturated to the precision this dataset supports; connectome $\leftrightarrow$ connectome reconstruction has a negligible nonlinear component. The tractography dead-end survives the most sensitive nonlinear probe we can construct.

N4 -- single-modality residual-boost

N4cog (cognition): NULL final = OOF-BR template + KernelRidge(residual). Delta(final - template) was +/-0.008, mixed sign -- KR residual adds ~0. Every tractography rep stayed -0.13 to -0.25 below the bv+demo floor (unchanged from linear E5/N1). FC stayed above floor.

N4recon (reconstruction): TINY REAL SIGNAL final vs template (PLS), demeaned_pearson, 10-seed Wilcoxon:

direction	template	final	Delta	p
FC->SC	0.1355	0.1405	+0.0052	0.001
SC->FC	0.0847	0.0866	+0.0023	0.001
FC->r2t	0.1113	0.1169	+0.0058	0.001
r2t->FC	0.0493	0.0526	+0.0041	0.001
FC->r2t_corr	0.0582	0.0623	+0.0050	0.001
r2t_corr->FC	0.0362	0.0402	+0.0039	0.001

Every direction improves at $p=0.001$ -- a genuine nonlinear residual the OOF framing surfaces where plain nonlinear (N2) saw only mixed-sign noise. But the effect is ~+0.005, practically negligible (counts still beat bundles; gap not closed). Identifiability metrics flat except r2t_corr (top1 +0.013 $p=0.02$; avg_rank +0.008 $p=0.01$).

N5 -- multimodal sink residual-boost (cross-modal interactions) The one architecture that can see cross-modal conjunctions: all modalities in one feature space, nonlinear model free to cross them.

N5cog (cognition): NULL + combining hurts

target	bv+demo floor	FC alone	sink_linear	sink_residual
CogTotal	0.373	0.436	0.401	0.381
CogFluid	0.298	0.306	0.301	0.286
CogCrystal	0.349	0.452	0.414	0.415

- **sink_linear** [FC||SC||r2t||bv||demo] is **-0.05 to -0.08 below FC alone**: adding SC/tractography/demographics to FC *dilutes* cognition prediction (extra dims, no new signal). FC alone is the best single predictor.
- **sink_residual - sink_linear** = -0.002 to -0.020 (Wilcoxon p~0.96-0.999 in the wrong direction): the nonlinear cross-modal residual does not help -- slightly hurts.
- **No FCxtractography interaction exists.** The conjunctive-modulator hypothesis (the most plausible way tractography could matter) is empty.

N5recon (reconstruction): NULL [SC||r2t||bv||demo] ->FC: sink_residual - sink_linear = -0.0023 (p=0.935). The tiny N4recon nonlinear sliver does not survive the multimodal sink.

Decision rules -- all returned null (except the negligible N4recon whisper)

test	rule	result
N4cog cognition unlock	tractography clears floor by >=0.02	NO (all below floor)
N4recon reconstruction gain	final - template >= 0.02 dp	NO (+0.005, though p=0.001)
N5cog cross-modal unlock	sink_residual - sink_linear >= 0.02, p<0.05	NO (-0.01, p>0.95 wrong way)
N5cog complementarity	sink_linear - FC >= 0.02	NO (-0.05 to -0.08, sink worse)
N5recon cross-modal recon	sink_residual - sink_linear >= 0.02	NO (-0.002)

Why this is the definitive version A clean null from OOF residual-boost means $y - \hat{y}_{linear}$ is **noise with respect to the source** -- not "the model couldn't find it," but "there is no learnable structure left." The linear model already extracted everything predictable at this n. Handing the nonlinear model the answer for free and tasking it only with the residual is the most sensitive probe possible; it found nothing for cognition, single-modality OR cross-modal.

The one exception (N4recon +0.005, p=0.001) is the honest nuance: connectome<->connectome reconstruction (same data type, dense edges) has a real but trivial nonlinear component; connectome->cognition (crossing into behavior) is dead-flat linear-saturated.

Caveats

- KernelRidge alpha/gamma fixed (median-heuristic gamma, alpha=1.0); decision margins wide vs effect sizes, so tuning won't flip the nulls.
- HGB excluded here (underperformed even on FC at n~683 in N1); KR is the trustworthy probe and the one used throughout N4/N5.
- n~683 train: the "linear ceiling" is partly a sample-size ceiling. A much larger cohort could in principle surface faint nonlinearity; nothing here suggests it would be practically meaningful.
- Single parcellation (Glasser), 10 family-aware seeds.

Files

- n4_recon_results.csv/_summary.csv, n4_cog_results.csv/_summary.csv
- n5_cog_results.csv/_summary.csv, n5_recon_results.csv/_summary.csv
- *_output.txt per experiment; residual_synthesis.csv
- scripts: n4_*, n5_*, _residual.py (OOF template + additive-residual helpers, single + multimodal-block variants), run_one.sbatch
- design: DESIGN_residual_learning.md

7.10 Sample-Size Scaling Check

7.10.1 N6 -- data-scaling learning curve: model-ceiling vs data-limited

Run: 2026-06-14, job 10782789, 10 seeds, multimodal sink, train subsampled at $n = 100/200/400/\text{full}(\sim 683)$, test fixed+full at every n . Question: does the nonlinear gap (residual-boost final - linear template) grow with n ? If yes -> signal is data-limited (bigger cohort indicated). If flat -> structural ceiling (more data won't help).

TL;DR MODEL-CEILING / STRUCTURAL. The nonlinear gap does not grow into positive territory with n in either task. Cognition is flat-and-slightly-negative at every n ; reconstruction is negative at every n and merely creeps toward zero from below as data grows (an overfitting-penalty effect, not signal). Meanwhile the *linear* performance climbs with n in both tasks -- the learning curve works, more data helps the linear model, and nonlinearity adds nothing at any sample size. **A bigger cohort is NOT indicated for nonlinearity; the linear-saturation conclusion is definitive.**

Results

Cognition (CogTotal), nonlinear gap vs n

n	median linear	median final	gap	Wilcoxon $p(\text{gap}>0)$
100	0.302	0.290	-0.0087	1.00
200	0.327	0.324	-0.0152	0.99
400	0.394	0.380	-0.0119	0.999
~ 683	0.391	0.375	-0.0128	0.99

gap-vs- n Spearman $\rho=+0.05$, $p=0.77$ -> **flat**. Gap is ≤ 0 at every n (nonlinear mildly *hurts*). Linear grows 0.30->0.39 with n . No nonlinear signal at any size.

Reconstruction (sink->FC), nonlinear gap vs n

n	median linear	median final	gap	Wilcoxon $p(\text{gap}>0)$
100	0.0412	0.0353	-0.0054	0.999
200	0.0670	0.0655	-0.0019	0.88
400	0.0940	0.0931	-0.0015	0.99
~ 683	0.1098	0.1097	-0.0009	0.98

gap-vs- n Spearman $\rho=+0.52$, $p=0.001$ -- **but this is NOT data-limited signal**. The gap is **negative at every n** and just rising *toward zero from below*; Wilcoxon confirms gap ≤ 0 everywhere ($p(\text{gap}>0) = 0.88-1.0$). The positive trend is the KernelRidge residual's **overfitting penalty shrinking as data grows** -- it asymptotes *at* zero, not above it. It never reaches the +0.02 "matters" threshold, never goes positive. Linear grows 0.04->0.11 with n .

(Note: full-train size is 682 for one seed, 683 for nine -- a family-split rounding artifact that splits the last column into two buckets; the single-seed 682 row is noise, ignore it. Plot: `n6_scaling_curve.png`.)

Why the auto-verdict was overridden The synthesizer's first pass flagged reconstruction as "DATA-LIMITED" on $\rho>0$ & $p<0.05$ alone. That's wrong: a negative gap creeping toward zero has a positive trend but is not emerging signal. Fixed the rule to require $\text{gap@max}_n > 0.005$ (actually positive and meaningful), which correctly returns MODEL-CEILING. Lesson: a positive *slope* on a *negative* gap is an overfitting penalty vanishing, not signal appearing -- always check the gap sign/level, not just the trend.

Interpretation

- **Cognition is linearly saturated, structurally.** Nonlinearity doesn't help at $n=100$ or $n=683$; the gap doesn't trend up. More subjects make the *linear* predictor better (0.30->0.39) but won't unlock nonlinear structure -- there is none to unlock at this representation/precision.

- **Reconstruction has no positive nonlinear component either**, at any n . The tiny nonlinear sliver seen in N4recon (+0.005 on full data, single-modality) does not appear in the multimodal sink and does not grow with n here; the residual penalty just converges to ~ 0 .
- **The bottleneck is information, not data or model capacity**. We've now shown the null is robust to: model class (N1-N3), the residual-boost head-start (N4), cross-modal interactions (N5), AND sample size (N6). Four independent ways of asking "is there nonlinear/extra signal," four nulls.

What this closes, and the one honest caveat Closes the "maybe a bigger model / harder tuning would find it" question: it wouldn't -- the gap doesn't grow with data, so it's not a capacity-or-sample-size ceiling, it's structural. The defensible headline: **the connectome->cognition relationship (and the FC->SC reconstruction) is linearly saturated; FC is the ceiling; richer tractography adds nothing -- across model classes, architectures, and sample sizes up to $n \sim 683$.**

Caveat: we tested to $n \sim 683$. We cannot rule out that $n \sim 10^4$ (UK Biobank scale) surfaces a tiny positive gap -- but our data shows convergence *to* zero, not growth *above* it, so there is no positive evidence for it. If one wanted to chase it, the move is a bigger cohort (and the expectation should be a very small effect), NOT a bigger model on these data.

Files

- `n6_scaling_curve.py`, `synthesize_scaling.py`, `_residual.py` (n -aware block helpers)
- `n6_scaling_results.csv`, `n6_scaling_summary.csv`, `scaling_synthesis.csv`
- `n6_scaling_curve.png` (gap-vs- n , both tasks)
- `n6_scaling_output.txt`

7.11 Family Mechanism Reproduction Notes

7.11.1 Family-Structure + Mechanism Grid (F6 / F7 / F8)

Second grid pass: replicates the family-structure (F6), predictor/identifier-tradeoff (F7), and PC-mechanism (F8) findings on **both parcellations x 10 frozen seeds**, so they stop being Glasser-only single-run results. Faithful ports of the notebooks (canonical source): `model_overviews/crossmodal_pca_pls_closed_form_overview.ipynb` STEP 8.1-8.3 (F6/F7) and `further_exploration/depth1*.ipynb` (F8). The notebook is authoritative -- if a helper drifts, fix the notebook first.

Status

- **F6 + F7 -- BUILT & VALIDATED.** Aggregation reproduces the notebook's `aggregate_auc.csv` bit-for-bit (AUC err $1e-16$, perm-p err 0, bootstrap CI err 6e-8). See tests below.
- **F8 -- BUILT & VALIDATED.** Port of `depth1_spectral_mechanism.ipynb` + `depth1.1_pc_stability_and_confounds.ipynb` (per-PC FC->PC R^2 , per-PC family AUC, sex+bv confound, PC1 residualization, cross-seed PC alignment, PC3 network enrichment via `data/atlas_info/<parc>dseg_reformatted.csv`; network col Glasser=community_yeo / 4S456=network_label). Localization math reproduces the notebook's saved seed-0 `sc_pc_loadings.npy` exactly (energy 0.630678, interhemi base 0.501393, top-50 edge region-mapping 50/50).

Files

- `_fm_common.py` -- shared layer. Lazy data import (`_data()`; needs torch/Torch) so the pure aggregation helpers run anywhere. Copies the 4 notebook-only helpers verbatim (`zscore_by_unrelated`, `perm_p_auc`, `bootstrap_auc`, `fdr_bh`); reuses `_setup`'s family helpers + predictors. `build_family_variants` / `pair_sims_for_seed` / `aggregate_family`.
- `run_f6_family.py` --`parc <P>` --`seed <S>` -- one unit: 8 variants -> per-pair sims -> per-unit npz. Self-checks Glasser/seed0 pair counts against the notebook (MZ=33,DZ=13,sib=125,unrel=171).

- `finalize_fm.py` -- pool all seeds per parc -> `outputs/family_auc.csv`; **regression guard** asserts Glasser == notebook aggregate within $1e-6$.
- `run_fm_unit.sbatch` -- array 0-19 (2 parc x 10 seeds), 24G/4h/8CPU (same profile as main grid).
- `tests/` -- see below.

Files (F8)

- `_f8_common.py` -- PC-mechanism helpers (parc-aware atlas, per-PC AUC/ R^2 , cosine alignment, energy, localization/enrichment) + `compute_seed()` (data layer, runs on Torch).
- `run_f8_pcmech.py` --parc <P> --seed <S> -- one unit -> per-unit npz; self-checks Glasser/seed0 against the notebook (pair counts, FC->PC R^2 , confound, PC3 AUC).
- `finalize_f8.py` -- cross-seed alignment + confound + PC3 localization/enrichment + verdict -> `outputs/f8_{per_pc,stability,pc3_localization,pc3_enrichment_agg}.csv`; Glasser regression guard.
- `run_f8_unit.sbatch` / `finalize_f8.sbatch` / `submit_f8.sh` -- same light profile as F6/F7.

Tests (run locally, no torch / no connectome data)

```
python reproduction/family_mechanism/tests/test_aggregation_matches_notebook.py # F6/F7 vs notebook CSV
python reproduction/family_mechanism/tests/test_helpers.py # F6/F7 helper correctness
python reproduction/family_mechanism/tests/test_f8_localization_matches_notebook.py # F8 vs saved seed-0 loadings
python reproduction/family_mechanism/tests/test_f8_helpers.py # F8 helper correctness
```

`test_aggregation_matches_notebook.py` feeds the notebook's own per-seed .npz (the expensive connectome-derived pair sims) into our aggregation and asserts we reproduce `aggregate_auc.csv` exactly -- the strongest off-cluster "matches the notebook" check. `test_helpers.py` pins the 4 copied helpers against reference implementations (FDR vs statsmodels, AUC vs sklearn, etc.).

Run on Torch One command -- submits the 20-unit array and chains finalize via `--dependency=afterok` (finalize runs only if every unit succeeds). Nothing to poll; watch the sentinels.

```
cd /scratch/ans9868/Conn2Conn/reproduction/family_mechanism
bash submit_fm.sh
### progress (passive, NO queue):
ls sentinels/DONE_fm_*.sentinel 2>/dev/null | grep -v finalize | wc -l # /20
test -f sentinels/DONE_fm_finalize.sentinel && echo FINALIZE DONE
cat logs/fm_finalize.txt ; column -s, -t outputs/family_auc.csv | head
```

Dedicated scripts (light profile -- F6/F7 has no KR sweep / no downstream / no F8):

- `run_fm_unit.sbatch` -- array 0-19 %10, **16G / 8 CPU / 1 h** (peak RSS ~4-8 GB).
- `finalize_fm.sbatch` -- single task, 8G / 30 min, runs `finalize_fm.py` in-container.
- `submit_fm.sh` -- submits both with the dependency wired.

Estimated runtime: ~3 min/unit (Glasser) / ~5 min (4S456) -- anchored on the notebook's `seed_*.npz` timestamps. ~15-20 min wall at %10 (2 waves) + ~3-5 min finalize, plus SLURM queue. 1 h time limit is generous (worst unit ~5 min); zero timeout risk.

What F6/F7 produce `outputs/family_auc.csv`: per (parcellation, variant, relation) AUC + bootstrap CI + perm-p + FDR.

- **F6** headline: `pred_SC_resid_bvdemo` sibling AUC (notebook Glasser = 0.810) vs `bvdemo_to_SC` baseline (0.563) -- family-specific wiring beyond shared anatomy.
- **F7** is read off the same table: `pred_SC_raw` separates siblings (0.680) but combined `pred_SC` collapses to chance (0.505, n.s.) -- reconstruct OR identify, not both. (The downstream-cognition half of F7 -- combined `pred_SC` -> cognition lift -- is the one new input to add to the main downstream grid; not required for the family-collapse evidence.)

8 Closing Interpretation

Taken together, these checks support a narrow but strong conclusion. FC-SC translation is reproducible and directional, but reconstruction accuracy is not a sufficient proxy for downstream cognitive utility. The null is not explained by the PCA reduction axis, obvious nonlinear model capacity, richer tractography features, sample-size trends within the tested regime, FC-side measurement noise, or hard grid leakage. The constructive exception is family signal: predicted connectomes can preserve identity/family information when the objective selects for it, but that does not imply cognition transfer.

8.1 Source Manifest

- **spine:** preprint/ supplement.md
- **reproduction_findings:** reproduction/ reports/ reproduction_findings.md
- **exploration_findings:** reproduction/ exploration/ FINDINGS_EXPLORATION.md
- **discrepancy:** reproduction/ exploration/ DISCREPANCY_RESOLUTION.md
- **family_readme:** reproduction/ family_mechanism/ README.md
- **preprocessing:** notebooks-FC_to_SC-experimental/ sanity_checks/ preprocessing_check/ findings.md
- **noise:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ findings_noise.md
- **tract_check:** notebooks-FC_to_SC-experimental/ sanity_checks/ tract_check/ findings.md
- **nonlinear:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ findings_nonlinear.md
- **residual:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ findings_residual.md
- **scaling:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ findings_scaling.md
- **tractography:** notebooks-FC_to_SC-experimental/ tractography_predict/ findings.md
- **Reduction-axis summary:** notebooks-FC_to_SC-experimental/ sanity_checks/ preprocessing_check/ reduction_axis_summary.csv
- **Reduction-axis all seed ratios:** notebooks-FC_to_SC-experimental/ sanity_checks/ preprocessing_check/ reduction_axis_synthesis.csv
- **FC reliability ceiling:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ a_reliability_ceiling.csv
- **FC per-subject reliability summary:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ g_per_subject_summary.csv
- **Reliability filtering summary:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ h_reliability_filtered_summary.csv
- **Noise synthesis:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ noise_synthesis.csv
- **Tractography source representation:** notebooks-FC_to_SC-experimental/ tractography_predict/ e1_source_rep_results.csv
- **Tractography asymmetry summary:** notebooks-FC_to_SC-experimental/ tractography_predict/ e2_asymmetry_summary.csv
- **Tractography marginal summary:** notebooks-FC_to_SC-experimental/ tractography_predict/ e3_marginal_summary.csv
- **Tractography downstream summary:** notebooks-FC_to_SC-experimental/ tractography_predict/ e5_downstream_summary.csv
- **Nonlinear cognition summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n1_cognition_summary.csv
- **Nonlinear reconstruction summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n2_reconstruction_summary.csv
- **Nonlinear marginal summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n3_marginal_summary.csv
- **Residual cognition summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n4_cog_summary.csv
- **Residual reconstruction summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n4_recon_summary.csv
- **Sink cognition summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n5_cog_summary.csv

- **Sink reconstruction summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n5_recon_summary.csv
- **Scaling summary:** notebooks-FC_to_SC-experimental/ non-linear-sanity-check/ n6_scaling_summary.csv
- **Family AUC:** reproduction/ family_mechanism/ outputs/ family_auc.csv
- **F8 stability:** reproduction/ family_mechanism/ outputs/ f8_stability.csv
- **Downstream reproduction grid:** reproduction/ outputs/ downstream.csv
- **Reconstruction reproduction grid:** reproduction/ outputs/ reconstruction.csv
- **Leak verdict grid:** reproduction/ outputs/ leak_verdict.csv
- **Expected grid cells:** reproduction/ configs/ expected_cells.csv
- **Detailed tractography downstream:** notebooks-FC_to_SC-experimental/ tractography_predict/ e5_downstream_results.csv
- **Per-subject FC achieved-vs-ceiling:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ h_per_subject_achieved_vs_ceiling.csv
- **FC achieved-vs-ceiling correlations:** notebooks-FC_to_SC-experimental/ sanity_checks/ noise_sanity_check/ outputs/ h_correlations.csv